

Notes on QFT

PROBABILITY FLOOR

August 2026

Source: <https://github.com/probability569/qft-notes-schwartz>

“note to self, do this or im the biggest retard”

feyman likely

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Preface

todo

1 Math

1.1 Notation

- (Loose) Einstein Summation notation is used in which if an index appears twice in both top and bottom, such as $\partial_\mu \partial^\mu$, it is equivalent to summing over said indices such as $\sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
 - The reason that this is loose is due to the fact that if two indices appear twice in an expression but they are both on the bottom, it is implied that the reader should raise the second appearance of the index. The following is an example so the reader doesn't get confused: $\partial_\mu \partial_\mu = \partial_\mu \partial^\mu = \sum_{\mu \in \{0,1,2,3\}} \partial_\mu \partial^\mu$
- The *Dirac Delta Function* is defined as

$$\delta(x) \equiv \frac{d}{dx} H(x)$$

with

$$\int dx \delta(x) = 1$$

This is not, in fact, a function, due to its unorthodox definition. Instead, it is a distribution (it can be defined as the limit of several different distributions, such as the *normal distribution*, a *uniform distribution*, etc.).

- $\vec{x} = x^i, i \in \{1, 2, 3\}$ (spatial components of x)
- The *d'Alembertian* operator $\square \equiv \partial_\mu \partial^\mu$

This operator is useful in *Lagrangians* and the *Klein-Gordon equation*.

- The *Laplacian* operator $\triangle \equiv \partial_i \partial^i$ where $i \in \{1, 2, 3\}$
- Natural Units are used in which $\hbar = c = 1$
- Both Sided Derivative: $\phi^* \overleftrightarrow{\partial}_\mu \phi \equiv (\phi^* \partial_\mu \phi - \phi \partial_\mu \phi^*)$
- Feynman Slash: $\not{p} \equiv p_\mu \gamma^\mu$
- Pauli matrices

$$\sigma^0 = \begin{pmatrix} 1 & \\ & 1 \end{pmatrix}, \sigma^1 = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix}, \sigma^2 = \begin{pmatrix} & -i \\ i & \end{pmatrix}, \sigma^3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

and $\sigma^\mu \equiv (\sigma^0, \vec{\sigma}), \bar{\sigma}^\mu \equiv (\sigma^0, -\vec{\sigma})$

- $$\gamma^\mu = \begin{pmatrix} & \sigma^\mu \\ \bar{\sigma}^\mu & \end{pmatrix}$$

in the Weyl representation

- $\bar{\psi} \equiv \psi^\dagger \gamma^0$ where ψ is a Dirac spinor
- Left Handed Derivative

$$\begin{aligned} & \bar{\psi}(-i\vec{\partial} - m) \\ &= -i\partial_\mu \bar{\psi} \gamma^\mu - m\bar{\psi} \end{aligned}$$

(Derivative acts on the left)

- $k_\mu^2 \equiv k_\mu k^\mu$
- ψ^α left handed Weyl spinor
- $\tilde{\psi}_{\dot{\alpha}}$ right handed Weyl spinor
- $\psi = \begin{pmatrix} \psi^\alpha \\ \tilde{\chi}_{\dot{\beta}} \end{pmatrix}$ Dirac spinor (Representation of $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$)

All of these expressions regarding spinors will be explained or will quickly become evident in Section 2, due to the complexity of their structure.

1.2 Integrals that are useful

- $\int dx f(x) \delta(x^2 - a^2) = \frac{1}{2|a|} (f(a) + f(-a))$
- $\int d^3k \theta(k^0) \delta(k_\mu^2 - m^2) = \frac{1}{2\omega_k} [\theta(\omega_k) + \theta(-\omega_k)] = \frac{1}{2\omega_k}$

where $\omega_k = \sqrt{\vec{k}^2 + m^2}$

- $\int \frac{d^4k}{(2\pi)^4} \exp(ik(x-y)) = \delta^4(x-y)$
- $\int \frac{d^Nk}{(2\pi)^N} \exp(ik(x-y)) = \delta^N(x-y)$
- $\int dx \exp\left(-\frac{1}{2}ax^2 + Jx\right) = \left(\frac{2\pi}{a}\right)^{\frac{1}{2}} \exp\left(\frac{J^2}{2a}\right)$

1.3 Group Theory

Definition 1.3.1. A Group G is a set of elements $\{g_i\} \in G$ and a rule $g_i \times g_j = g_k$ giving how they “multiply.”

1.3.1 Group Theory Axioms

- For each element $g_i \in G$ and $g_j \in G$, their product $g_i \times g_j = g_k \in G$ is an element of G
- Products must be Associative:

$$(g_i \times g_j) \times g_k = g_i \times (g_j \times g_k)$$

- There must be an Identity:

$$\mathbb{1} \times g_i = g_i \times \mathbb{1} = g_i$$

- Elements have an Inverse:

$$g_i^{-1} \times g_i = \mathbb{1}$$

Definition 1.3.1.1 (Representation). An embedding of g_i as operators in a vector space

- Matrices are finite-dimensional representations of a group

Definition 1.3.1.2 (Faithful Representation). Each element in a group gets its own matrix

1.3.2 Lie Groups

- A Group G whose elements can be written as $g_i = \exp(ic_i^g \lambda_i)$ for some number c_i^g and some matrix λ_i (known as a *generator*)
- In an algebra, you can add and multiply
- In a Group, you have to follow the previously mentioned rules (just multiplication is allowed)

Definition 1.3.2.1 (Lie Group). A *Lie Group* is a type of group with an infinite number of elements and a finite number of generators.

Definition 1.3.2.2 (Lie Algebra). The Generators of a *Lie group* form a *Lie Algebra*.

Example 1.3.2.3 (Lie Algebras).

- QED : $U(1)$ (unitarity)
- Weak: $SU(2)$ (special unitarity)
- Strong: $SU(3)$
- Lorent Group: $O(1, 3)$ (orthogonal and preserving (1,3) metric) (see §2.2)

2 Scalar Field Theory

- Scalar Field Theory is a theory of quantum fields. It is mostly used as a toy theory to base other theories off of, since out of the many *many* particles that exist (dozens counting baryons), there is an entirety of **1** scalar particle (the Higgs boson). Therefore, Scalar Field Theory primarily is only useful to base other models of the universe off of it as opposed to modeling the universe after it.

3 Classical Field Theory

3.1 Classical Field Theory

- The idea behind **classical field theory** is to utilize the idea of Lagrangians that will be presented in *Lagrangians* and in *Classical Symmetries* to obtain a “toy theory” without the quantum effects that will later appear such as Loops in diagrams, etc. In practice, this means that it is essentially the *Lagrangian Formalism* of Classical Mechanics where *position* and *momentum* coordinates (p, q) are replaced by fields (ϕ, π) , as seen before in *Lagrangians* without having been explicitly stated.
- Classical Field Theory, much like *Classical Mechanics*, functions as the unquantized equivalent to Quantum Field Theory. Quantization does not require much to achieve as simply putting quantization conditions on the operators being used achieves it with relative ease.

Example 3.1.1 (Klein Gordon Equation).

- A very standard equation for a field in Classical Field theory is the *Klein Gordon Equation*. (This, in fact, describes a *free* field in which there are no interactions, allowing for a very good “toy theory” to use and potentially apply later).

$$(\square + m^2)\phi = 0$$

or

$$(\partial^2 + m^2)\phi = 0$$

is the *Klein Gordon Equation*.

3.2 Lagrangians

3.2.1 Lagrangians

Definition 3.2.1.1 (Lagrangian). The *Lagrangian* (Density) $\mathcal{L}$, similar to the *Hamiltonian* H (or $\mathcal{H}$ for Hamiltonian density, defined as $H = \int d^3x \mathcal{H}$) as they both can be used to describe the evolution of a system. In Field Theories, densities are used more commonly. The Density of either quantity, as stated before, is a more useful quantity in Field theories due to Lorentz Invariance and other such things.

Definition 3.2.1.2 (Action). Action S is defined as

$$S = \int dt L = \int d^4x \mathcal{L}$$

which should be familiar from studying classical mechanics (Lagrangian mechanics specifically). This quantity is minimized in order to obtain the evolution of a system (**Equations of Motion**).

Theorem 3.2.1.3 (EULER LAGRANGE EQUATIONS). *In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu \phi]$, the system must satisfy the Euler Lagrange Equations:*

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = 0$$

A differential equation which ends up being extremely useful in physics for describing the time evolution of a system.

Proof. We aim to equate the variation of the action S with respect to the field ϕ to zero. That is, we require

$$\frac{\delta S}{\delta \phi} = 0$$

Note that

$$S = \int dt L = \int d^4x \mathcal{L}[\phi, \partial_\mu \phi]$$

in order to obtain δS , we must perform a transformation $\phi \rightarrow \phi + \delta \phi$ in the Lagrangian. This provides

$$\delta S = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta (\partial_\mu \phi) \right)$$

Via the product rule and the fact that infinitesimal variations commute with partial derivatives¹, we have that

$$\begin{aligned} \delta S &= \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) \\ \delta S &= \int d^4x \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta \phi - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) \right) \\ \delta S &= \int d^4x \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \delta \phi + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \delta \phi \right) \right) \end{aligned}$$

¹Look into variational calculus if you want to learn more about this

Now we need to prove that the final term (the one without the shared common factor of $\delta\phi$) vanishes as our integration bounds become infinite (as we are integrating over all spacetime of course). Since this term is a total derivative of some quantity that we will define to be

$$A^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi$$

(Something similar with the form of $\frac{A^\mu}{\delta\alpha}$ is a physically important quantity but this A^μ is not of any significance besides being useful for math). With the derivative being of course $\partial_\mu A^\mu$. The integral of this quantity is

$$\int d^4x \partial_\mu A^\mu$$

and by the **4 dimensional divergence theorem**, we obtain

$$\int_V d^4x \partial_\mu A^\mu = \oint_{\partial V} d^3\sigma_\mu A^\mu$$

This must vanish, given that the $\delta\phi$ is forced to vanish at the boundaries by the principle of least action. Therefore, we have that

$$\frac{\delta S}{\delta\phi} = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0$$

For the integral to vanish, the integrand must vanish, therefore

$$\frac{\partial \mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = 0$$

□

Theorem 3.2.1.4 (EXTENDED EULER LAGRANGE EQUATIONS). *In order to minimize the action of a system with Lagrangian Density $\mathcal{L}[\phi, \partial_\mu \phi, \partial_\nu \partial_\mu \phi, \dots]$, the system must satisfy the **Extended Euler Lagrange Equations**:*

$$\frac{\partial \mathcal{L}}{\partial\phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \dots = 0$$

(This was actually a problem in the Schwartz QFT textbook, so if you are reading that, potentially skip this part.)

Proof. We begin in a similar way to the proof to the Euler-Lagrange equations.

$$\delta S = \int d^4x \left(\frac{\partial \mathcal{L}}{\partial\phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) + \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \partial_\mu \phi) + \dots \right)$$

We then utilize the product rule (or the reverse of it) to obtain

$$\begin{aligned}
\partial_\nu \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \right) &= \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \\
&+ \partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\mu \phi) \\
&+ \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \phi) \\
&+ \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \partial_\mu \phi)
\end{aligned}$$

We can rewrite $\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\nu \phi)$ and $\partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta(\partial_\mu \phi)$ via similar methods and we can continue to do so with more terms forming a pattern of the n th (with n partial derivatives) term has a coefficient of $(-1)^n$ (this method similar to the *principle of inclusion-exclusion*)

$$\begin{aligned}
\delta S &= \int d^4x \left(\left(\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \dots \right) \delta \phi \right. \\
&+ \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta \phi \right) + \partial_\mu \left(\partial_\nu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \right) + \partial_\nu \left(\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} \delta \phi \right) \\
&\left. + \partial_\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \partial_\nu \phi)} \delta \phi \right) + \dots \right)
\end{aligned}$$

Every term besides those with the shared factor of $\delta \phi$ disappears due to similar reasons mentioned in the proof of the *Euler-Lagrange equations* (i.e. they are total derivatives). and therefore, since we want $\frac{\delta S}{\delta \phi} = 0$, we have that

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} + \partial_\nu \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\nu \partial_\mu \phi)} - \dots = 0$$

□

Example 3.2.1.5 (Lagrangian). An example of a Lagrangian that can apply to some general scalar field ϕ^2 is the Lagrangian

$$\mathcal{L} = \frac{1}{2} (\partial_\mu \phi)(\partial_\mu \phi) - V[\phi]$$

This would give us the equation of motion

²That is, it doesn't apply universally, but just in this circumstance

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} &= 0 \\ -V'[\phi] - \square \phi &= 0\end{aligned}$$

so

$$(\square + V'[\phi])\phi = 0$$

which is related to the *Klein-Gordon equation* $(\square + m^2)\phi = 0$.

- One can split the terms of a Lagrangian $\mathcal{L}$ into *Kinetic* terms and *Interacting* terms.
 - *Kinetic* terms of a Lagrangian are ones which have at most 2 fields in it

$$- \mathcal{L}_{\text{kin}} \supset \frac{1}{2}\phi\square\phi, \bar{\psi}\not{\partial}\psi, \frac{1}{4}F_{\mu\nu}^2, \frac{1}{2}m^2\phi^2, \dots$$

- Interacting terms are terms with more than 2 fields

$$\mathcal{L}_{\text{int}} \supset \lambda\phi^3, g\bar{\psi}A\psi, g\partial_\mu\phi A_\mu\phi^*, \dots$$

Remark. Interacting terms in the lagrangian all have a factor λ or g associated with them. This is due to the fact that it is a coupling constant related to the interaction.

3.3 Classical Symmetries

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3.3.1 Classical Symmetries

3.3.1.1 Noether's Theorem

- This theorem is the core in the utilizing symmetries in classical and non-classical field theory.

Theorem 3.3.1.1.1 (NOETHER'S THEOREM). *If a Lagrangian has a continuous symmetry (i.e. it is parameterizable by some continuous α), there exists a conserved current J^μ associated with said symmetry when the equations of motion of the Lagrangian are satisfied, i.e. the **Euler-Lagrange** equations. In other words, symmetries imply conservation laws.*

Proof. The Proof for *Noether's Theorem* is fairly short and clever.

We define $\mathcal{L}$ as the Lagrangian of the system, ϕ_n as the field(s) involved in the system, and α as the parameter of the continuous symmetry.

Similar to the derivation of the *Euler-Lagrange* equations,

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left(\left(\frac{\partial \mathcal{L}}{\partial \phi_n} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \right) \frac{\delta \phi_n}{\delta \alpha} + \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} \right)$$

This expression must be equivalent to zero due to α parametrizing a symmetry (i.e. the Lagrangian is invariant wrt some variation in the parameter). It is necessary to define a

$$J^\mu \equiv \sum_n \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha}$$

Since we are only considering the case in which the equations of motion hold true, (i.e. the *Euler-Lagrange equations* are true), we have that

$$\partial_\mu J^\mu = \partial_\mu \sum_n \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = \sum_n \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) \frac{\delta \phi_n}{\delta \alpha} = 0$$

Therefore due to some continuous symmetry parametrized by some α , there exists a conserved current J^μ . $\square$

- The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ is the conserved current for spacetime translation.

Definition 3.3.1.1.2 (Energy Momentum Tensor). The *Energy-Momentum Tensor* $\mathcal{T}_{\mu\nu}$ for some Lagrangian $\mathcal{L}[\phi_1, \partial_\mu \phi_1, \dots, \phi_n, \partial_\mu \phi_n]$ and some space-time translation $x^\nu \rightarrow x^\nu + \xi^\nu$ is defined as

$$\mathcal{T}_{\mu\nu} \equiv \sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L}$$

Derivation below.

Starting with the translation

$$\phi(x) \rightarrow \phi(x + \xi) = \phi(x) + \partial_\nu \xi^\nu \phi(x) + \dots$$

This provides $\frac{\delta \phi}{\delta \xi^\nu} = \partial_\nu \phi$ which applies universally regardless of the field ϕ (and in fact regardless of the physical object, as it applies to the Lagrangian $\mathcal{L}$, a consequence of the fact that $\delta S = 0$).

We then have that

$$\begin{aligned}
\frac{\delta \mathcal{L}}{\delta \xi^\nu} &= \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \xi^\nu} \right) \\
&= \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n \right) \\
\partial_\nu \mathcal{L} &= \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n \right)
\end{aligned}$$

We then have that

$$\partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \right) = 0$$

Where we then obtain

$$\mathcal{T}_{\mu\nu} \equiv \left(\sum_n \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L} \right)$$

and $\partial_\mu \mathcal{T}_{\mu\nu} = 0$.

- Energy density:

$$\mathcal{E} = \mathcal{T}_{00} = \sum_n \frac{\partial \mathcal{L}}{\partial (\dot{\phi}_n)} \dot{\phi}_n - \mathcal{L}$$

- $\dot{\phi}_n = \partial_t \phi_n$
- In the case of Lagrangians with $\dot{\phi} = \pi = \frac{\partial \mathcal{L}}{\partial \phi}$, $\mathcal{E} = \mathcal{H}$ due to the *Legendre Transform*

Definition 3.3.1.1.3 (4-Momentum). In terms of the energy-momentum tensor, 4-momentum p^μ is defined as

$$p^\nu \equiv \int d^3x \mathcal{T}^{0\nu} = (E, \vec{p})$$

3.3.1.2 More information on currents

- A current can refer to a Noether current (such as the ones previously seen)
- They can be external, such as a moving electron in a wire
- They can be sources for fields
 - They are never fields with kinetic terms
- They can be placeholders in Lagrangians

Example 3.3.1.2.1 (Currents in Lagrangians). In

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tilde{\phi}\square\phi - ieA_\mu(\tilde{\phi}\vec{\partial}_\mu\phi),$$

we can write it as

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \tilde{\phi}\square\phi - A_\mu J^\mu$$

- Clearly, you can see how currents are used to represent quantities in lagrangians that we do not care for.

3.4 Propagators

3.5 Propagators (TODO)

3.5.1 Green's Functions

Definition 3.5.1.1 (Green's Functions). A Green's function is a function of the form A^{-1} where A is an operator. A standard way to evaluate these green function operators is via a Fourier transform into k -space¹. This is a type of *propagator*.

Definition 3.5.1.2 (Propagator). Defines or is used to define how a field propagates via applying the propagator Π to some other field or current. Can be used perturbatively in order to describe more complicated fields.

- An example of a propagator is the 2-point Green's function $\Pi = -\frac{1}{\square}$ which is defined by

$$\square_x \Pi(x, y) = -\delta^4(x - y)$$

- With $\square_x = g^{\mu\nu} \frac{\partial}{\partial x^\mu} \frac{\partial}{\partial x^\nu}$ This gives the solution (via the Fourier transform), that

$$\Pi(x, y) = \int \frac{d^4 k}{(2\pi)^4} e^{ik(x-y)} \frac{1}{k^2}$$

.

===Fourier Transform of Operators Under *Fourier Transforms*, certain operators are effectively interchangeable with some counterpart that often simplifies calculations. An example of this is how $\Pi(x, y)$ is interchangeable with $\frac{1}{k^2}$.

Example 3.5.1.3 (Delta Functions). For an instance of this occurrence, we can utilize delta functions.

¹ k -space is effectively momentum space due to the relation $p = \hbar k$ and the fact that we are using natural units

The Fourier transform of any delta function is

$$\tilde{\delta}(\vec{k}) = 1$$

$$\Rightarrow \delta(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} e^{i\vec{k}\cdot\vec{x}}$$

$$\begin{aligned} \Delta^n \delta^3(\vec{x}) &= \int \frac{d^3k}{(2\pi)^3} \Delta^n e^{i\vec{k}\cdot\vec{x}} \\ &= \int \frac{d^3k}{(2\pi)^3} (-\vec{k}^2)^n e^{i\vec{k}\cdot\vec{x}} \end{aligned}$$

- $\Delta^n \leftrightarrow (-\vec{k}^2)^n$ (this means that in a *Fourier Transform*, these are interchangeable).
- Additionally, we can do the same for the relativistic $\square = \partial_\mu \partial^\mu$

$$\square^n(\delta^4(x)) = \int \frac{d^4k}{(2\pi)^4} \square^n e^{ik\cdot x} = \int \frac{d^4k}{(2\pi)^4} (-k^2)^n e^{ik\cdot x}$$

- Have that $[\widetilde{\square^n \delta^4}](k) = (-k^2)^n$ and $[\widetilde{\Delta^n \delta^3}](k) = (-\vec{k}^2)^n$

3.5.2 Perturbation Theory

Definition 3.5.2.1 (Perturbation Theory). This topic is introduced in Quantum mechanics (and often before-hand) where we consider an initial state that we can solve and then higher order terms that we continue to be able to solve. Perturbation theory only works when the leading term is the “dominant” force in play. Often, we can utilize an initial term which is *non-interacting* and extra terms that are *interacting*.

Example 3.5.2.2 (Gravity). This example was provided from the Schwartz textbook² We start with the Lagrangian

$$\mathcal{L} = -\frac{1}{2}h\square h + \frac{1}{3}\lambda h^3 + Jh$$

where J is some *current* and h is our field. From the *Euler-Lagrange equations*, our equation of motion is

$$\square h - \lambda h^2 - J = 0$$

First order approx ($\lambda = 0$): $h_0 = \frac{1}{\square}J$. We let $h = h_0 + h_1$.

$$\square(h_0 + h_1) - \lambda(h_0 + h_1)^2 - J = 0$$

²This is the primary textbook that I am learning Quantum Field Theory through, as seen in the credits section.

implies

$$\square h_1 = \lambda h_0^2 + \mathcal{O}(\lambda^2)$$

$$h = \frac{1}{\square} J + \lambda \frac{1}{\square} \left(\left(\frac{1}{\square} J \right) \left(\frac{1}{\square} J \right) \right) + \mathcal{O}(\lambda^2)$$

which we now have in terms of our Green's Functions.

In fact, something being defined as $\frac{1}{\square} J$ is not that uncommon, as it also happens with the Electromagnetic field.

3.5.3 OFPT

- For **Old Fashioned Perturbation Theory** we utilize perturbation by setting the Hamiltonian to

$$H = H_0 + V$$

- The idea is that we have some $|\phi\rangle$ and some $|\psi\rangle$ such that

$$H_0|\phi\rangle = E|\phi\rangle$$

and

$$H|\psi\rangle = E|\psi\rangle$$

- From this we can derive

$$|\psi\rangle = |\phi\rangle + \frac{1}{E - H_0} V|\phi\rangle,$$

the *Lippmann-Schwinger* equation

- This allows us to define the propagator³

$$\Pi_{\text{LS}} \equiv \frac{1}{E - H_0}$$

- We define an operator T as

$$V|\psi\rangle = T|\phi\rangle$$

$$|\psi\rangle = |\phi\rangle + \Pi_{\text{LS}} T|\phi\rangle$$

$$V|\psi\rangle = V|\phi\rangle + V\Pi_{\text{LS}} T|\phi\rangle$$

³In fact, this is not well-defined, but the limit

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{E - H_0 + i\varepsilon}$$

is.

- Note that $V|\psi\rangle = T|\phi\rangle$ so

$$T|\phi\rangle = V|\phi\rangle + V\Pi_{\text{LS}}T|\phi\rangle = (V + V\Pi_{\text{LS}}T)|\phi\rangle.$$

(This holds when contracted with any eigenstate of H_0 , $|\phi_j\rangle$)

- This implies that $T = V + V\Pi_{\text{LS}}T$.
- Recursion provides

$$T = V + V\Pi_{\text{LS}}V + V\Pi_{\text{LS}}V\Pi_{\text{LS}}V + \dots$$

If we want a transition amplitude for this, we want to calculate

$$\langle f|T|i\rangle = \langle f|V|i\rangle + \langle f|V\Pi_{\text{LS}}V|i\rangle + \dots$$

- We can insert the eigenstates within this equation due to the fact that

$$\sum_j |\phi_j\rangle\langle\phi_j| = \mathbb{1}$$

This provides

$$\langle\phi_f|T|\phi_i\rangle = \langle\phi_f|V|\phi_i\rangle + \langle\phi_f|V\Pi_{\text{LS}}|\phi_j\rangle\langle\phi_j|V|\phi_i\rangle + \dots$$

- In an alternative/more convenient notation,

$$T_{fi} = V_{fi} + \left(\sum_j\right) V_{fj}\Pi_{\text{LS}}(j)V_{ji} + \left(\sum_j\right) V_{fi}\Pi_{\text{LS}}(j)V_{jk}\Pi_{\text{LS}}(k)V_{ki} + \dots$$

Where $T_{ij} = \langle\phi_i|T|\phi_j\rangle$, $V_{ij} = \langle\phi_i|V|\phi_j\rangle$, $\Pi_{\text{LS}}(j) = \frac{1}{E - E_j}$ with $E_f = E_i$ and the sums tend to be omitted

- We can with a certain particle state $|i\rangle$ and end with a final certain particle state $|f\rangle$ and use *OFPT* to calculate what happens in-between such states. An example of this is the case of an electron scattering off an electron.

Example 3.5.3.1 (Electron Scattering). This is an example from the textbook that I was reading which allows for the derivation of the *Advanced* and *Retarded* propagators.

- The transition matrix element is

$$T_{fi} = V_{fi} + \sum_n V_{fn} \frac{1}{E_i - E_n} V_{ni} + \dots$$

- For electron scattering, the initial and final states are $|i\rangle = |\psi_e^1\psi_e^2\rangle$ and $\langle f| = \langle\psi_e^3\psi_e^4|$
- Due to the fact that Coulomb's law is replaced by photon exchange we need a photon field $\phi(x)$.

- This provides

$$V = \frac{1}{2}e \int d^3x \psi_e(x) \phi(x) \psi_e(x)$$

with the $\frac{1}{2}$ from spin and the e as some coupling constant (which happens to be the electron charge).

- This interaction turns a $\vec{p}_1$ electron into a $\vec{p}_3$ electron and a $\vec{p}_\gamma$ photon.
- Since both $|i\rangle$ and $|f\rangle$ have 2 electrons and no photons, $V_{ij} = 0$

3.5.3.1 Retarded and Advanced Propagators

- This gets split into two states. (propagated by the *Retarded* Propagator and the *Advanced* Propagator)

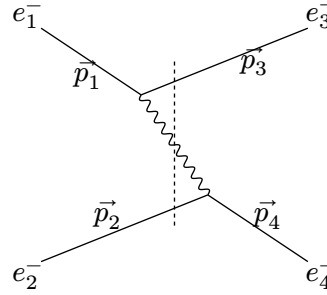


Figure 3.1: Retarded state

- The propagators that this creates are foundational to quantum field theory, (the advanced and retarded ones), which is the primary reason behind this methodology.
- In the Retarded state, electrons interact via the *Retarded Propagator*.
- We have that

$$|n\rangle = |\psi_e^3 \phi^\gamma \psi_e^2\rangle$$

- Which implies that

$$\begin{aligned} V_{ni}^R &= \langle \psi_e^3 \phi^\gamma \psi_e^2 | V | \psi_e^1 \psi_e^2 \rangle \\ &= \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle \langle \psi_e^2 | \psi_e^2 \rangle \\ &= \langle \psi_e^3 \phi^\gamma | V | \psi_e^1 \rangle \end{aligned}$$

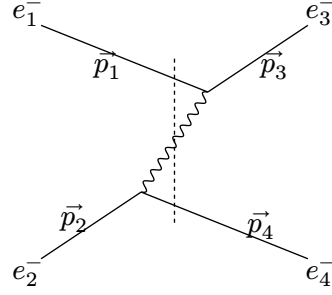


Figure 3.2: Advanced State

- For the *advanced state*, a similar result is derived

$$\text{▶ } V_{ni}^A = \langle \psi_e^4 \varphi^\gamma | V | \psi_e^2 \rangle$$

3.5.3.2 Evaluating The Propagators

$$\bullet \quad V_{ni}^R = \langle \psi_e^3 \varphi^\gamma | V | \psi_e^1 \rangle = \frac{e}{2} \int d^3x \langle \psi_e^3 \varphi^\gamma | \psi_e(x) \phi(x) \psi_e(x) | \psi_e^1 \rangle$$

- ▶ Free field (like the photon and electron fields inside of the integral)

$$\langle \phi^\gamma | \phi(x) | 0 \rangle = e^{-i\vec{p} \cdot \vec{x}}$$

- This implies:

$$\begin{aligned} V_{ni}^R &= e \int d^3x e^{i(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) \cdot \vec{x}} \\ &= e(2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_3) \end{aligned}$$

- ▶ This is similar to other matrix elements

- First order:

$$T_{fi} = \int d^3\vec{p}_\gamma (2\pi)^3 \delta^3(\vec{p}_1 - \vec{p}_2 - \vec{p}_\gamma) (2\pi)^3 \delta^3(\vec{p}_2 - \vec{p}_4 + \vec{p}_\gamma) \frac{e^2}{E_i - E_n}$$

- ▶ Energy is not conserved in such intermediate states (otherwise $E_i = E_n$ would blow u

3.5.3.3 Energy Calculations

- We want to calculate the energy of the states

3.5.3.4 Here are some denotations:

- Momenta of particles are

$$\vec{p}_1, \vec{p}_2, \vec{p}_3, \vec{p}_4, \vec{p}_\gamma$$

- The energy of each respective particle is

3.5.4 Interactions

3.6 More Information on the S-matrix

- The S -matrix describes the interaction between particles in the system, and as previously seen, it is defined as

$$\langle f|S|i\rangle = \mathbb{1} + (2\pi)^4 \delta^4(p_f - p_i) \mathcal{M}$$

- This S-matrix is the primary goal of most calculations in quantum field theory (as it describes the interactions of the particles)

3.6.1 LSZ Reduction

- The way to obtain the S-matrix from the *Time-Ordered Product* involves the **LSZ Reduction Formula** which is extremely important, and allows for the derivation of the extremely simple momentum-space Feynman Rules, which you will see later.
- The LSZ reduction formula is as follows:

$$\begin{aligned} \langle f|S|i\rangle = & \left[i \int d^4x_1 e^{-ip_1x_1} (\square + m^2) \right] \dots \left[i \int d^4x_n e^{ip_nx_n} (\square + m^2) \right] \\ & \times \langle \Omega | T \phi(x_1) \phi(x_2) \phi(x_3) \dots \phi(x_n) | \Omega \rangle \end{aligned}$$

Definition 3.6.1.1 (LSZ Reduction). The *LSZ Reduction* is a formula providing S -matrix elements from the time ordered product (in the ground state for the interacting theory Ω).

- Additionally, this formula can allow for the calculation of transfer matrix elements $\mathcal{M}$, but it is more convenient to utilize Feynman diagrams derived from this formulation.
 - The cross sections are calculatable from this as well
- This formula effectively deletes terms in the time ordered product except for those with poles of $\frac{1}{p^2 - m^2}$
- The ground state in the interacting theory, $|\Omega\rangle$ is a very inconvenient way to calculate, due to lack of knowledge about the ground state

3.6.1.1 Correlation Functions

- A **correlation function** $\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle$ is the time ordered product of fields which appears in the LSZ reduction formula after the integrals
- This is specifically used in the derivation of the LSZ reduction formula

3.6.1.2 Derivation of LSZ Reduction

- Suppose that the initial state $|i\rangle$ and the final state $|f\rangle$ are of the form

$$|i\rangle = \sqrt{2\omega_1}\sqrt{2\omega_2}a_{p_1}^\dagger(-\infty)a_{p_2}^\dagger(\infty)|\Omega\rangle$$

and

$$|f\rangle = \sqrt{2\omega_1}\cdots\sqrt{2\omega_j}a_{p_3}^\dagger(\infty)\cdots a_{p_j}^\dagger(\infty)|\Omega\rangle$$

- Additionally there is the constraint that $|i\rangle \neq |f\rangle$ due to scattering happening.
- The S-matrix element is then

$$\langle f|S|i\rangle = 2^{\frac{j}{2}}\sqrt{\omega_1\cdots\omega_n} \times \langle \Omega|a_{p_3}(\infty)\cdots a_{p_j}(\infty)a_{p_1}^\dagger(-\infty)a_{p_2}^\dagger(-\infty)|\Omega\rangle$$

- Now it is desirable to convert this to scalar fields, ϕ
- From quantization, they are defined by

$$\phi = \int \frac{d^3p}{(2\pi)^3} (a_p(t)e^{-ipx} + a_p^\dagger(t)e^{ipx})$$

3.6.1.2.1 Heisenberg picture to obtain operators

- Consider the Heisenberg picture
 - $a_p(t) = e^{iH(t-t_0)}a_p(t_0)e^{-iH(t-t_0)}$ and $\phi(x) = e^{iH(t-t_0)}\phi(x, t_0)e^{-iH(t-t_0)}$ via unitary time evolution
 - The only useful thing to know about $\phi(x, t)$ and $a_p(t)$ is

$$\langle \Omega|\phi(x, t = \pm\infty)|p\rangle = Ce^{-ip\cdot x}$$

- A useful identity:

$$i \int d^4x e^{ipx} (\square + m^2)\phi(x) = \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)]$$

- Proof:

$$\begin{aligned} i \int d^4x (\square + m^2)\phi(x) &= i \int d^4x e^{ipx} (\partial_t^2 - \vec{\partial}_x^2 + m^2)\phi \\ &= i \int d^4x e^{ipx} (\partial_t^2 + \vec{p}^2 + m^2)\phi(x) \\ &= i \int d^4x e^{ipx} (\partial_t^2 + \omega_p^2)\phi(x) \end{aligned}$$

- Identity for proof:

$$\begin{aligned}
& \partial_t [e^{ipx} (i\partial_t + \omega_p) \phi(x)] \\
&= [i\omega_p e^{ipx} (i\partial_t + \omega_p) + e^{ipx} (i\partial_t^2 + \omega_p \partial_t^2)] \phi(x) \\
&= ie^{ipx} (\partial_t^2 + \omega^2) \phi(x)
\end{aligned}$$

► Putting this into the integral,

$$\begin{aligned}
i \int d^4x e^{ipx} (\square + m^2) \phi(x) &= \int d^4x \partial_t [e^{ipx} (i\partial_t + \omega_p) \phi(x)] \\
&= \int dt \partial_t \left[e^{i\omega_p t} \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \phi(x) \right]
\end{aligned}$$

– This is true for all $\phi(x)$

► Quantum Field Case:

$$\begin{aligned}
& \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \phi(x) \\
&= \int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \\
&\quad \times \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} (a_k(t) e^{-ikx} a_k^\dagger(t) e^{ikx}) \\
&= \int \frac{d^3k}{(2\pi)^3} \int d^3x \left(\left(\frac{\omega_k + \omega_p}{\sqrt{2\omega_k}} \right) a_k(t) e^{-ikx} e^{-i\mathbf{p}\cdot\mathbf{x}} \right. \\
&\quad \left. + \left(\frac{-\omega_k + \omega_p}{\sqrt{2\omega_k}} \right) a_k^\dagger(t) e^{ikx} e^{-i\mathbf{p}\cdot\mathbf{x}} \right)
\end{aligned}$$

Note that

- $\partial_t a_k(t) = 0$ at $t = \pm\infty$
- The d^3x integral enforces $\omega_k = \omega_p$ via delta functions
- Therefore, we get

$$\int d^3x e^{-i\mathbf{p}\cdot\mathbf{x}} (i\partial_t + \omega_p) \phi(x) = \sqrt{2\omega_p} a_p(t) e^{-i\omega_p t}$$

• This implies

$$\begin{aligned}
i \int d^4x e^{ipx} (\square + m^2) \phi(x) &= \int dt \partial_t \left((e^{i\omega_p t}) (\sqrt{2\omega_p} a_p(t) e^{-i\omega_p t}) \right) \\
&= \sqrt{2\omega_p} [a_p(\infty) - a_p(-\infty)]
\end{aligned}$$

as desired

3.6.1.2.2 S matrix element computing

- Now we can compute

$$\begin{aligned} \langle f|S|i\rangle &= 2^{\frac{j}{2}} \sqrt{\omega_1 \omega_2 \dots \omega_j} \\ &\times \langle \Omega | a_{p_3}(\infty) \dots a_{p_j}(\infty) a_{p_1}^\dagger(-\infty) a_{p_2}^\dagger(-\infty) | \Omega \rangle \end{aligned}$$

Definition 3.6.1.2.2.1 (Time Ordered Product).

- The time ordered product $T\mathcal{O}(x_1)\mathcal{O}(x_2)\dots\mathcal{O}(x_n)$ picks up the operators $\mathcal{O}$ and literally places them so that they are correctly ordered in time. This is the only thing that the operator does, as it keeps the products there.
- Notice that

$$\begin{aligned} \langle f|S|i\rangle &= \sqrt{2^j \omega_1 \dots \omega_j} \langle \Omega | T[a_{p_3}(\infty) - a_{p_3}(-\infty)] \dots [a_{p_j}(\infty) - a_{p_j}(-\infty)] \\ &\times [a_{p_1}^\dagger(-\infty) - a_{p_1}^\dagger(\infty)] [a_{p_2}^\dagger(-\infty) - a_{p_2}^\dagger(\infty)] | \Omega \rangle \end{aligned}$$

- Therefore, we have the LSZ Reduction Formula

$$\begin{aligned} \langle p_3 \dots p_j | S | p_1 p_2 \rangle &= \left[i \int d^4 x_1 e^{-ip_1 x_1} (\square_1 + m^2) \right] \\ &\times \dots \times \left[i \int d^4 x_j e^{ip_j x_j} (\square_j + m^2) \right] \\ &\times \langle \Omega | T \phi(x_1) \phi(x_2) \dots \phi(x_j) | \Omega \rangle \end{aligned}$$

3.6.1.3 LSZ For Operators

- The LSZ Reduction formula is relatively powerful, as it applies for all operators $\mathcal{O}_j$
- This implies that

$$\begin{aligned} \langle p_3 \dots p_j | S | p_1 p_2 \rangle &= \left[i \int d^4 x_1 e^{-ip_1 x_1} (\square_1 + m^2) \right] \\ &\times \dots \times \left[i \int d^4 x_j e^{ip_j x_j} (\square_j + m^2) \right] \\ &\times \langle \Omega | T \mathcal{O}_1(x_1) \dots \mathcal{O}_j(x_j) | \Omega \rangle \end{aligned}$$

3.7 Quantization and Fock

3.8 Scalar Field Theory

- Scalar Field Theory is fairly similar to Classical Field Theory with the exception that it is quantized via the methods that will be presented shortly. This ensures that it obeys the laws of quantum mechanics, while still being a

“Toy Theory” due to the simplicity of the objects involved in it ((potentially) complex fields ϕ as opposed to *vectors* A_μ or *spinors* ψ which are closer to real, physical particles).

3.8.1 Quantization

- As seen in ‘s Quantum Field Theory Notes, one could try to naively try to quantize the Schrodinger Equation (via the use of a series to have the equation be on-shell), which would bring rise to some problems

$$H|\psi\rangle = i\partial_t|\psi\rangle$$

$$H = \sqrt{\hat{p}^2 + m^2}$$

3.8.1.1 Harmonic Oscillator/First Quantization

- The standard Hamiltonian for a simple harmonic oscillator is

$$H = \frac{1}{2} \frac{p^2}{m} + \frac{1}{2} m\omega^2 x^2$$

. The relevance to quantization will become apparent.

Definition 3.8.1.1.1 (First Quantization). The idea behind first quantization is to apply the commutation relation $[\hat{x}, \hat{p}] = i$ to a system. This commutation relation is known as the *canonical commutation relation*, and it can be used to obtain quantum mechanics from a classical theory (which we are doing now).

- Utilizing a change of variables to a and $a^\dagger$, one can derive an alternative form for the Hamiltonian

$$a \equiv \sqrt{\frac{m\omega}{2}} \left(x + \frac{ip}{m\omega} \right), \quad a^\dagger \equiv \sqrt{\frac{m\omega}{2}} \left(x - \frac{ip}{m\omega} \right)$$

►
$$H = \omega \left(a^\dagger a + \frac{1}{2} \right)$$

is written in a very suggestive form. Such $a^\dagger$ and a can be thought of as “exciting” the harmonic oscillator, as you may know from quantum mechanics.

- The eigenstates of this Hamiltonian are eigenstates of $\hat{N} \equiv a^\dagger a$
- Notice that this hamiltonian’s eigenvalues (the energies of the eigenstates) are discrete. This is a result from quantum mechanics (or, rather, defining quantum mechanics). Such a result is the reason behind both the name *Quantum Mechanics* and the idea of *First Quantization*.
- $\hat{N}$ operator:
 - $\hat{N}|n\rangle = n|n\rangle$
 - $a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$

$$\triangleright a|n\rangle = \sqrt{n}|n-1\rangle$$

3.8.1.2 Heisenberg Picture

- $i\frac{d}{dt}a = [a, H] = \left[a, a\left(a^\dagger a + \frac{1}{2}\right) \right] = \omega(aa^\dagger a - a^\dagger aa) = \omega[a, a^\dagger]a = \omega a$
- Solution is

$$a(t) = e^{-i\omega t}a(0)$$

3.8.1.3 Relativistic Fields

- A common relativistic equation (heavy foreshadowing) for a (free) field is the equation

$$\square\phi = 0.$$

- Solution to this is

$$\phi(x, t) = \int \frac{d^3p}{(2\pi)^3} [a_p(t)e^{i\vec{p}\cdot\vec{x}} + \tilde{a}_p(t)e^{-i\vec{p}\cdot\vec{x}}]$$

which is written in a fairly suggestive manner

3.8.1.4 2nd Quantization

Definition 3.8.1.4.1 (Second Quantization). To obtain 2nd quantization (and hence quantum field theory), one can use an annihilation operator a_p and a creation operator $a_p^\dagger$ which exist per wave number as opposed to being universal since now, we utilize infinite harmonic oscillators which exist per p

- Free Hamiltonian is obtained via integration over these

$$H = \int \frac{d^3p}{(2\pi)^3} \omega_p \left(a_p^\dagger a_p + \frac{1}{2} \right)$$

with $\omega_p = |\vec{p}|$

- 2nd Quantization new things:
 - Each (of the infinitely many) point of momentum $\vec{p}$ gets its own harmonic oscillator (which we integrate over to obtain our result)
 - The n th excitation of the harmonic oscillator $\vec{p}$ as being a state with n particles

3.8.1.5 Fock Space

- Instead of Hilbert space $\mathcal{H}$, like in Quantum Mechanics/First Quantization, we have a *Fock Space* $\mathcal{F}$.
- A Fock space $\mathcal{F}$ is the direct sum of Hilbert spaces $\mathcal{H}_n$, that is,

$$\mathcal{F} = \bigoplus_n \mathcal{H}_n$$

- States in $\mathcal{H}_n$ are linear combinations of $\{|p_1^\mu \dots p_n^\mu\rangle\}$
 - Since we are using *Fock Space*, this means that there can be however many particles due to the direct sum over Hilbert Spaces

3.8.1.6 Momentum Notation

$p^0 = \sqrt{\vec{p}^2 + m^2}$ so $|p\rangle = |\vec{p}\rangle = |p^\mu\rangle$ due to no loss of generality (this only works if it is on-shell).

3.8.1.7 Field Expansion

- Previously, we noted that $[a, a^\dagger] = 1$. This generalizes to $[a_k, a_p^\dagger] = (2\pi)^3 \delta^3(\vec{p} - \vec{k})$
- $a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |p\rangle$
- $\langle 0|0\rangle = 1$ since it is the vacuum ground state
 - $\langle p|k\rangle = 2\sqrt{\omega_p \omega_k} \langle 0|a_p a_k^\dagger|0\rangle = 2\omega_p (2\pi)^3 \delta^3(\vec{p} - \vec{k})$
- Identity operator

$$\mathbb{1} = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_p} |p\rangle \langle p|$$

- Free field definition

$$\phi_0(\vec{x}) \equiv \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{i\vec{p}\cdot\vec{x}} + a_p^\dagger e^{-i\vec{p}\cdot\vec{x}})$$

which is why I said $\phi(\vec{x}, t)$ from 4.1.2 was written in a suggestive manner (although this is still technically just a definition)

3.8.1.8 Momentum state inner product with free field

$$\begin{aligned} \langle \vec{p} | \phi_0(\vec{x}) | 0 \rangle &= \langle 0 | \sqrt{2\omega_p} a_p \int \frac{d^3k}{(2\pi)^3} (a_k e^{i\vec{k}\cdot\vec{x}} + a_k^\dagger e^{-i\vec{k}\cdot\vec{x}}) | 0 \rangle \\ &= \int \frac{d^3k}{(2\pi)^3} \sqrt{\frac{\omega_p}{\omega_k}} \left[e^{i\vec{k}\cdot\vec{x}} \langle 0 | a_p a_k | 0 \rangle + e^{-i\vec{k}\cdot\vec{x}} \langle 0 | a_p a_k^\dagger | 0 \rangle \right] = e^{-i\vec{p}\cdot\vec{x}} \\ &= e^{-i\vec{p}\cdot\vec{x}} \end{aligned}$$

This is the expected result of $\langle \vec{p} | \vec{x} \rangle$, and so we conclude $\phi_0(\vec{x}) | 0 \rangle = |\vec{x}\rangle$

3.8.1.9 Time Dependence

- Because of Heisenberg's equations of motion, in the Heisenberg perspective, $a_p(t)$ and $a_p^\dagger(t)$ are time dependent
 - $a_p(t) = e^{-i\omega_p t} a_p, a_p^\dagger(t) = e^{i\omega_p t} a_p^\dagger$

- Heisenberg equations of motion

$$\begin{aligned}
[H_0, \phi_0(\vec{x}, t)] &= \int \frac{d^3p}{(2\pi)^3} \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left[\omega_p \left(a^\dagger_p a_p + \frac{1}{2} \right), a_k e^{-ikx} + a_k^\dagger e^{ikx} \right] \\
&= \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[-\omega_p a_p e^{-ipx} + \omega_p a_p^\dagger e^{ipx} \right] \\
&= -i\partial_t \phi_0(\vec{x}, t)
\end{aligned}$$

- Should have that, for any field ϕ ,

$$[H, \phi(\vec{x}, t)] = i\partial_t \phi(\vec{x}, t)$$

$$- \quad \phi(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \left[a_p(t) e^{-ipx} + a_p^\dagger(t) e^{ipx} \right]$$

is the solution to the *Heisenberg equations of motion* for ϕ , our field which does not necessarily have to be free in this case

4 Cross Sections

- Cross sections are how physicists analyze collider data (such as the exabytes of LHC data that is collected every year)

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2} |\mathcal{M}|^2$$

and the Born approximation

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{Born}} = \frac{m_e^2}{4\pi^2} [\tilde{V}(\vec{k})]^2$$

- Cross sections used to make sense of particle accelerator data
- Cross sectional area:

$$\begin{aligned}
\sigma &= \frac{\# \text{ particles}}{\text{time} \times \text{number density} \times \text{velocity of beam}} \\
&= \frac{1}{T} \frac{1}{\Phi} N, \Phi = \# \text{ density} \times \text{beam vel.} = \frac{1}{V} \times |\vec{v}|
\end{aligned}$$

- More useful/natural is differential cross section

$$\frac{d\sigma}{d\Omega}$$

for some angle/change in angle Ω .

- Info about shape of object with this

4.0.0.1 QM

- Generalize to Differential Cross Sections due to their changing based on angle in colliders
- In QM, there is a probability of scattering

$$P = \frac{N}{N_{\text{inc}}}$$

with N_{inc} number of incident particles

- Cross section

$$d\sigma = \frac{1}{T} \frac{1}{\Phi} dP$$

(normalize flux)

- Differential scattering number

$$dN = L \times d\sigma$$

- L is defined as the *Luminosity* through this equation

4.0.1 Cross Section to Matrix Element

- $2 \rightarrow N$ Particle Interaction:

$$p_1 + p_2 \rightarrow \{p_j\}$$

-

$$\Phi = \frac{|\vec{v}_1 - \vec{v}_2|}{V}$$

$$\Rightarrow d\sigma = \frac{V}{T} \frac{1}{|\vec{v}_1 - \vec{v}_2|} dP$$

- Have that

$$dP = \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle} d\Pi$$

- with

$$d\Pi = \prod_j \frac{N}{(2\pi)^3} d^3p_j$$

-

$$\int d\Pi = 1$$

since

$$\int \frac{dP}{(2\pi)} = \frac{1}{V}$$

- $[a_p, a_q^\dagger] = (2\pi)^3 \delta^3(p - q)$

- Implies that $\langle p|p \rangle = (2\pi)^3 \delta^3(0)$
- Normally $\delta^3(0)$ is infinite but

$$(2\pi)^3 \delta^3(p) = \int_V d^3x e^{i\vec{p} \cdot \vec{x}}$$

$$\implies \delta^3(0) = \frac{V}{(2\pi)^3}$$

$$\text{and } \delta^4(0) = \frac{TV}{(2\pi)^4}$$

- $\langle p|p \rangle = 2\omega_p V = 2E_p V$

- Since

$$|i\rangle = |p_1\rangle |p_2\rangle$$

and

$$|f\rangle = \prod_j |p_j\rangle$$

- It is implied that

$$\langle i|i \rangle = (2E_1 V)(2E_2 V)$$

- It is also implied that

$$\langle f|f \rangle = \prod_j 2E_j V$$

4.0.2 Actually Calculating the S-matrix (perturbatively)

- Notice that the *first order* of a S-matrix is just $\mathbb{1}$ since that is the case in which nothing happens
- Write $S = \mathbb{1} + i\mathcal{T}$
 - $\mathcal{T} \equiv$ **transfer matrix**
 - $\mathcal{T}$ implies momentum conservation since it vanishes unless $|i\rangle$ and $|f\rangle$ have the same 4-momentum
- $\mathcal{T} = (2\pi)^4 \delta^4(\Sigma p) \mathcal{M}$

4.0.2.1 Matrix Elements

- $\langle f|S - \mathbb{1}|i \rangle = i(2\pi)^4 \delta^4(\Sigma P) \langle f|\mathcal{M}|i \rangle$
 - $|\langle f|S|i \rangle|^2 = \delta^4(0) \delta^4(\Sigma p) (2\pi)^8 |\langle f|\mathcal{M}|i \rangle|^2$

$$= \delta^4(\Sigma p) TV (2\pi)^4 |\mathcal{M}|^2$$

4.0.3 Lorentz Invariant Phase Space

- We denote the differential $d\Pi_{\text{LIPS}}$ as being defined as the product over final momenta differentials times delta function factors, that is

$$d\Pi_{\text{LIPS}} \equiv \prod_{\text{final } j} \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_{p_j}} (2\pi)^4 \delta^4(\Sigma p)$$

- Utilizing this, (as well as the derived S-matrix elements from earlier) we can write the differential probability dp via

$$\begin{aligned} dP &= \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle} d\Pi = \frac{(2\pi)^4 \delta^4(\Sigma p) \times TV}{2E_1 V \times 2E_2 V} \\ &\times \frac{1}{\Pi_j(2E_j V)} |\mathcal{M}|^2 \prod_j \frac{d^3 p_j}{(2\pi)^3} V \\ &= T \frac{1}{2E_1 \times 2E_2} |\mathcal{M}|^2 d\Pi_{\text{LIPS}} \end{aligned}$$

- This easily gives us the cross section for $2 \rightarrow j$ decay:

$$d\sigma = \frac{1}{(2E_1)(2E_2)|\vec{v}_1 - \vec{v}_2|} |\mathcal{M}|^2 d\Pi_{\text{LIPS}}$$

4.0.4 Particle Decay Rate

- This is another useful quantity in analyzing accelerator data denoted by $d\Gamma$
- The definition for this decay rate is $d\Gamma \equiv \frac{1}{T} dP$ (probability of decay in certain amount of time)
 - Specifically it is the probability that a particle of momentum p_1 decays into particles of momenta p_j in a time T
 - S-matrix theory cannot be used to describe these decays (?)
- $d\Gamma = \frac{1}{2E_1} |\mathcal{M}|^2 d\Pi_{\text{LIPS}}$

4.0.5 Cross Section formula

- Consider $2 \rightarrow 2$ Scattering
 - This means that the momenta are $p_1 + p_2 \rightarrow p_3 + p_4$
- In the center of mass frame, the net momentum for each is zero. Therefore $p_1 = -p_2, p_3 = -p_4$ which implies that $E_1 + E_2 = E_3 + E_4 \equiv E_{\text{CM}}$
- Then, for the LIPS differential, we get that

$$d\Pi_{\text{LIPS}} = (2\pi)^4 \delta^4(\Sigma p) \frac{d^3 p_3}{(2\pi)^3} \frac{1}{2E_3} \frac{d^3 p_4}{(2\pi)^3} \frac{1}{2E_4}$$

- This is equivalent to the integral

$$d\Pi_{\text{LIPS}} = \frac{1}{16\pi^2} d\Omega \int dp_f \frac{p_f^2}{E_3} \frac{1}{E_4} \delta(E_3 + E_4 - E_{\text{CM}})$$

(I don't understand how this is true)

- Where $p_f = |\mathbf{p}_3| = |\mathbf{p}_4|$ and $E_3 = \sqrt{m_3^2 + p_f^2}$, $E_4 = \sqrt{m_4^2 + p_f^2}$
- Notice that the energies are dependent on the final momentum p_f , and so this integral is in fact, not trivial.

- Now we conduct a change of variables from p_f to $x(p_f) \equiv E_3(p_f) + E_4(p_f) - E_{\text{CM}}$
- Calculating the change of variables, we get

$$\frac{dx}{dp_f} = \frac{d}{dp_f}(E_3 + E_4 - E_{\text{CM}}) = \frac{p_f}{E_3} + \frac{p_f}{E_4} = \frac{E_3 + E_4}{E_3 E_4} p_f$$

- We also have that

$$E_3 + E_4 = E_{\text{CM}}$$

- Then,

$$\begin{aligned} d\Pi_{\text{LIPS}} &= \frac{1}{16\pi^2} d\Omega \int_{m_3+m_4-E_{\text{CM}}}^{\infty} dx \frac{p_f}{E_{\text{CM}}} \delta(x) \\ &= \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{\text{CM}}} \theta(E_{\text{CM}} - m_3 - m_4) \end{aligned}$$

where $\theta(x)$ is the Heaviside step function due to the integral of the delta function.

- Putting this all together, we obtain that

$$d\sigma = \frac{1}{(2E_1)(2E_2)|\mathbf{v}_1 - \mathbf{v}_2|} \frac{1}{16\pi^2} d\Omega \frac{p_f}{E_{\text{CM}}} |\mathcal{M}|^2 \theta(E_{\text{CM}} - m_3 - m_4)$$

- Using

$$|\mathbf{v}_1 - \mathbf{v}_2| = \left| \frac{|\mathbf{p}_1|}{E_1} + \frac{|\mathbf{p}_2|}{E_2} \right| = |\mathbf{p}_i| \frac{E_{\text{CM}}}{E_1 E_2}$$

, we obtain that, under the condition that the masses are equivalent,

$$\left(\frac{d\sigma}{d\Omega} \right)_{\text{CM}} = \frac{1}{64\pi^2 E_{\text{CM}}^2} |\mathcal{M}|^2$$

as desired, allowing for computation of cross sections for simple scattering based off of quantum field theory predictions for $|\mathcal{M}|^2$

4.0.6 Born Approximation

- The Born approximation is a special case where a scalar electron ϕ_e with mass m_e scatters off of a scalar proton ϕ_p with mass m_p
 - It is considered the nonrelativistic limit
- I will not derive the born approximation, however, it is given by the formula

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Born}} \equiv \frac{m_e^2}{4\pi^2} [\tilde{V}(\vec{k})]^2$$

where $\tilde{V}$ is the fourier transformed potential/the potential in fourier space.

- $$\tilde{V}(\vec{K}) = \int d^3x e^{-i\vec{k}\cdot\vec{x}} V(\vec{x})$$

5 Feynman Diagrams

5.0.1 Feynman Propagators

- We can define the feynman propagator $G_F(x, y)$ to be implicitly defined by

$$(\square_x + m^2)G_F(x, y) = -i\delta^4(x - y)$$

- This provides us, in position space, with

$$\begin{aligned} G_F(x, y) &= \int \frac{d^4p}{(2\pi)^4} \frac{i}{p^2 - m^2 + i\varepsilon} e^{ip\cdot(x-y)} \\ &\equiv \int \frac{d^4p}{(2\pi)^4} G_F(p) e^{-ipx + iy} \end{aligned}$$

- The momentum space Feynman Propagator $G_F(p)$ is defined as

$$G_F(p) \equiv \frac{i}{p^2 - m^2 + i\varepsilon}$$

- The Feynman propagator, which we will elaborate upon, is a very fundamental quantity that is useful in evaluating Feynman diagrams (to be more explicit, it could be one of the only ways to evaluate them).
 - In conclusion, Richard Feynman is the GOAT

Definition 5.0.1.1 (Feynman Propagator).

- The Feynman propagator $G_F(x, y)$ or $D_F(x, y)$ or $G_F(x - y)$ or D_{xy} depending on the situation is defined as the time ordered product of free fields.
 - That is,

$$G_F(x, y) \equiv \langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle$$

$$= \lim_{\varepsilon \rightarrow 0} \int \frac{d^4 p}{(2\pi)^4} \frac{1}{k^2 - m^2 + i\varepsilon} e^{ik(x-y)}$$

Definition 5.0.1.2 (Feynman Diagram). A Feynman diagram is something that is used to represent particle interactions, or simplifications of them which are summed. generally to calculate the LSZ reduction for matrix elements. These matrix elements are ultimately used to calculate cross sections as seen in *Cross Sections*.

Definition 5.0.1.3 (Feynman Rules). Feynman rules are a set of rules for a theory which is used to evaluate Feynman diagrams.

- A standard set of Feynman rules that can generally apply everywhere are the following

General Position Space Feynman Rules

- At each vertex, a factor of the interaction coupling is given
- All lines are integrated over $d^3 x$ after applying the Feynman propagator
- The LSZ reduction formula is applied to convert this to an S-matrix element
- Additional factors may be added if they are incoming/outgoing electrons/photons due to polarization and spin

General Momentum Space Feynman Rules (Arguably much more useful :

- Each vertex of a Feynman Diagram gets a factor of whatever the interaction coupling is.
- Vertices can only occur at points that satisfy interactions in the Lagrangian¹
- For each edge whose momentum is known (that does not connect to an external vertex), a factor of the propagator for the theory Π is gained - For each edge whose momentum k is unknown (i.e., a loop), a factor of Π is gained and must be integrated over

$$\frac{d^4 k}{(2\pi)^4}.$$

- The final product gives $i\mathcal{M}$ for the diagram
- Additional factors may be added if they are incoming/outgoing electrons/photons due to polarization and spin

5.0.2 Wick's Theorem and Feynman Diagram Derivation

- Before diagrams, it is useful to do some calculations

¹For instance, if $\mathcal{L}_{\text{int}} = \frac{g}{3!} \phi^3$, only interactions which involve fields intersecting at three points are included in the set of tree-level diagram

- There are two different ways to derive Feynman rules for a theory, but the most common two are through the use of lagrangians and through the use of 2nd quantization

5.0.2.1 Lagrangian Method

- Consider Free Fields with

$$\begin{aligned} [\phi(\mathbf{x}, t), \phi(\mathbf{x}', t)] &= 0 \\ [\phi(\mathbf{x}, t), \partial_t \phi(\mathbf{x}', t)] &= i\hbar \delta^3(\mathbf{x} - \mathbf{x}') \\ (\square + m^2)\phi &= 0 \end{aligned}$$

- We only know how to evaluate the free theory correlation functions at this point

$$\langle 0 | T \phi(x) \phi(x') | 0 \rangle$$

Theorem 5.0.2.1.1 (SCHWINGER-DYSON EQUATIONS).

- *An equation that will appear later via the use of Feynman path integrals*
- *The Schwinger Dyson Equations state that*

$$\begin{aligned} (\square + m^2) \langle \phi_x \phi_1 \phi_2 \dots \phi_n \rangle &= \langle \mathcal{L}'_{\text{int}}[\phi_x] \phi_1 \dots \phi_n \rangle \\ -i\hbar \sum_j \delta^4(x - x_j) \langle \phi_1 \dots \phi_{j-1} \phi_{j+1} \dots \phi_n \rangle \end{aligned}$$

- *This holds if*

$$\mathcal{L} = -\frac{1}{2}(\square + m^2)\phi + \mathcal{L}_{\text{int}}[\phi]$$

- *They differentiate classical mechanics and quantum mechanics via the $\hbar$ term.*

Proof Schwinger-Dyson Equation.

- This should barely constitute a proof but nevertheless I have to call it something.
- Consider

$$\partial_t \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle$$

- This is equal to

$$\begin{aligned} &\partial_t [\langle \Omega | \phi(x) \phi(x') | \Omega \rangle \Theta(t - t') + \langle \Omega | \phi(x') \phi(x) | \Omega \rangle \Theta(t' - t)] \\ &= \langle \Omega | T \partial_t \phi(x) \phi(x') | \Omega \rangle + \langle \Omega | \phi(x) \phi(x') | \Omega \rangle \partial_t \Theta(t - t') \\ &\quad + \langle \Omega | \phi(x') \phi(x) | \Omega \rangle \partial_t \Theta(t' - t) \end{aligned}$$

- Using the fact that $\partial_T \Theta(t) = \delta(t)$, we obtain that the above expression is equal to

$$\langle \Omega | T \partial_t \phi(x) \phi(x') | \Omega \rangle + \delta(t - t') \langle \Omega | [\phi(x), \phi(x')] | \Omega \rangle$$

- Taking the second derivative, we obtain

$$\begin{aligned} \partial_t^2 \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle &= \langle \Omega | T \partial_t^2 \phi(x) \phi(x') | \Omega \rangle \\ &+ \delta(t - t') \langle \Omega | [\partial_t \phi(x), \phi(x')] | \Omega \rangle \end{aligned}$$

- Since the delta function forces time equivalence ($t = t'$) in the second term, we have that

$$[\partial_t \phi(x), \phi(x')] = -i\hbar \delta^3(\mathbf{x} - \mathbf{x}')$$

and therefore additionally have that

$$\partial_t^2 \langle \Omega | T \phi(x) \phi(x') | \Omega \rangle = \langle \Omega | T \partial_t^2 \phi(x) \phi(x') | \Omega \rangle - i\hbar \delta^4(x - x')$$

Chevron Notation

- As you may have seen earlier, we have introduced notation $\langle \phi \rangle$ in order to make notation more condensed. As you may be able to guess, $\langle \phi \rangle \equiv \langle \Omega | \phi | \Omega \rangle$
- Therefore, the equation with $\square + m^2$ follows and implies our Feynman propagator,

$$(\square_x + m^2) G_F(x, y) = -i\hbar \delta^4(x - y)$$

- That is,

$$(\square + m^2) \langle \phi(x) \phi(x') \rangle = \langle (\square + m^2) \phi(x) \phi(x') \rangle - i\hbar \delta^4(x - x')$$

- Applying Euler-Lagrange equations gives

$$\langle \mathcal{L}'_{\text{int}[\phi]} \phi(x) \phi(x') \rangle = i\hbar \delta^4(x - x')$$

- Generalization provides the final equation,

$$(\square_x + m^2) \phi = \langle \mathcal{L}'_{\text{int}} \phi_x \phi_1 \dots \phi_n \rangle - i\hbar \sum_j \langle \phi_1 \dots \phi_j \phi_{j+1} \dots \phi_n \rangle$$

□

- Now we can simply evaluate the two point functions
- For instance,

$$\langle \phi_1 \phi_2 \rangle = \int d^4 x \delta_{x1} \langle \phi_x \phi_2 \rangle$$

(this is simply using the identity that $\int d^4 x \delta(x) f(x) = f(0)$).

- The above expression is then equivalent to

$$i \int d^4 x (\square_x D_{x1}) \langle \phi_x \phi_2 \rangle = i \int d^4 x D_{x1} \square_x \langle \phi_x \phi_2 \rangle$$

- Some condensed notation was used where $D_{ab} = G_F(x_a, x_b)$ and $\delta_{xb} = \delta(x - x_b)$
- In a free theory $\square_x \langle \phi_x \phi_Y \rangle = -i \delta_{xy}$, and so

$$\langle \phi_1 \phi_2 \rangle = \int d^4 x D_{x1} \delta_{x2} = D_{12}$$

which is not very interesting (it is literally a particle going from point x to point y without anything happening)

5.0.2.2 Wick's Theorem/Contractions/QM Derivation

- Let us move to an interacting theory ($\mathcal{L}_{\text{int}} \neq 0$)

5.0.2.2.1 Consider the Heisenberg picture

- We have the Heisenberg equations of motion $i \partial_t \phi(x) = [\phi, H]$
 - These necessarily imply that $\phi(x, t)$ evolves via an operator S . i.e.

$$\phi(x, t) = S(t, t_0)^\dagger \phi(x) S(t, t_0)$$

, noting that S is the S-matrix.

- S satisfies $i \partial_t S(t, t_0) = H(t) S(t, t_0)$ (the schrodinger equation)
- In the Heisenberg picture, we do not need the explicit hamiltonian, just the time dependent part
 - This is because the states are time independent and the Hamiltonian is not
- $H(t) = H_0 + V(t)$

5.0.2.2.2 Interaction Picture

- In the interaction picture, fields only evolve via H_0
- Free fields:

$$\phi_0(x, t) = e^{iH_0(t-t_0)} \phi(x) e^{-iH_0(t-t_0)}$$

- This is equivalent to the plane wave solution for scalar fields. That is,

$$\phi_0(x, t) = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} (a_p e^{-ipx} + a_p^\dagger e^{ipx})$$

- In order to get the real field for this, we have that

$$\begin{aligned}\phi(\mathbf{x}, t) &= S^\dagger(t, t_0) e^{-iH(t-t_0)} \phi_0(\mathbf{x}, t) e^{iH_0(t-t_0)} S(t, t_0) \\ &\equiv U^\dagger(t, t_0) \phi_0(\mathbf{x}, t) U(t, t_0)\end{aligned}$$

- This $U(t, t_0) \equiv e^{iH_0(t-t_0)} S(t, t_0)$ is unitary time evolution operator that we can utilize

- It relates the Heisenberg picture to free fields at some time t

- Notice that it satisfies (a form of) the Schrodinger equation. Namely,

$$\begin{aligned}i\partial_t U(t, t_0) &= -e^{iH_0(t-t_0)} H_0 S(t, t_0) + e^{iH_0(t-t_0)} H(t) S(t, t_0) \\ &= e^{iH_0(t-t_0)} [-H_0 + H(t)] e^{-iH_0(t-t_0)} e^{iH_0(t-t_0)} S(t, t_0) \\ &= H_I(t) U(t, t_0)\end{aligned}$$

with $H_I(t) \equiv e^{iH_0(t-t_0)} V(t) e^{-iH_0(t-t_0)}$

- This $H_I(t)$ is the Heisenberg interaction/potential in the interacting picture
- The solution to this is

$$U(t, t_0) = T \exp \left(-i \int_{t_0}^t H_I(t') dt' \right)$$

which is known as the **Dyson Equation**

5.0.2.3 More Feynman rules derivation

- Consider the operator

$$U_{12} \equiv U(t_1, t_2) = T \exp \left(-i \int_{t_1}^{t_2} dt' H_I(t') \right)$$

- In this notation, the later time is on the left
- We have the following identities:
 - $U_{21} U_{12} = 1$
 - $U_{21}^{-1} = U_{21}^\dagger = U_{12}$
 - For $t_1 < t_2 < t_3$, $U_{32} U_{21} = U_{31}$
 - $U_{32} U_{12} = U_{32}$
- Since $\phi(x_1) = \phi(\mathbf{x}_1, t) = U_{10}^\dagger \phi_0(\mathbf{x}_1, t) U_{10} = U_{01} \phi_0(\mathbf{x}_1, t) U_{10}$

5.0.3 Vacuum Matrix Elements

- We want to evolve the ground state $|\Omega\rangle$ to a time t_0 such that the interacting picture and the free picture are equivalent

- In the interacting picture, states $|\Omega\rangle$ evolve via $S(t, t_0)$ and are annihilated by $a_p(t)$ at $t = -\infty$
- In the heisenberg picture, $a_p(t)$ evolves as $S(t, t_0)^\dagger a_p(t_0) S(t, t_0)$ and annihilates $|\Omega\rangle$ at $t = -\infty$
- In the free theory, $|0\rangle$ is annihilated by a_p
 - a_p evolves only via phase rotation
 - $a_p(t_0) e^{iH_0(t-t_0)} |0\rangle = 0$ at $t = -\infty$ (annihilation)
- We must have that $e^{iH_0(t-t_0)} |0\rangle$ and $S(t, t_0) |\Omega\rangle$ must be proportional
 - Then,

$$|\Omega\rangle = \mathcal{N}_i \lim_{t \rightarrow -\infty} S^\dagger(t, t_0) e^{iH_0(t-t_0)} |0\rangle = \mathcal{N}_i U_{0, -\infty} |0\rangle$$

- $\mathcal{N}_i$ is just a number

- Similarly, for the final time,

$$\langle \Omega | = \mathcal{N}_f \langle 0 | U_{\infty, 0}$$

for some $\mathcal{N}_f$

- Then, for our time ordered product, assuming that $t_1 > t_2 > \dots t_n$ (WLOG), we have that

$$\begin{aligned} \langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle &= \langle \Omega | \phi(x_1) \dots \phi(x_n) | \Omega \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | U_{\infty 0} U_{01} \phi_0(x_1) U_{10} U_{02} \phi_0(x_2) U_{20} \dots U_{0n} \phi_0(x_n) U_{n0} U_{0-\infty} | 0 \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | U_{\infty 1} \phi(x_1) U_{12} \phi(x_2) U_{23} \dots U_{(n-1)n} \phi_0(x_n) U_{n-\infty} | 0 \rangle \end{aligned}$$

- Now we can simply apply the time ordering, giving

$$\begin{aligned} &\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | T U_{\infty 1} \phi_0(x_1) U_{12} \phi_0(x_2) U_{23} \dots \phi_0(x_n) U_{n-\infty} | 0 \rangle \\ &= \mathcal{N}_i \mathcal{N}_f \langle 0 | T \phi_0(x_1) \dots \phi_0(x_n) U_{\infty, -\infty} | 0 \rangle \end{aligned}$$

- Additionally, $\langle \Omega | \Omega \rangle = 1$ which implies that, by normalization, $\mathcal{N}_i \mathcal{N}_f = \langle 0 | U_{\infty, -\infty} | 0 \rangle^{-1}$

Note: We have been taking $t \rightarrow \infty$ this whole time, which is incorrect. Utilizing $t \rightarrow \infty$ without any damping will result in oscillations. That is, if we insert the energies

$$e^{-iHt} |0\rangle \sum_n e^{iE_n t} |n\rangle \langle n| 0\rangle$$

, taking $t \rightarrow \infty$ yields e^{-itE} for an energy E which at large times causes unpredictable phase oscillations. Thus, we must use an analytic continuation by taking $t \rightarrow \infty - i\varepsilon$ (which would provide a dampening $e^{-\varepsilon E}$)

- We can finally have the time ordered product

$$\begin{aligned} & \langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle \\ &= \lim_{T \rightarrow \infty - i\varepsilon} \frac{\langle 0 | T \phi_0(x_1) \dots \phi_0(x_n) \exp \left[-i \int_{-T}^T dt H_I(t) \right] | 0 \rangle}{\langle 0 | T \exp \left[-i \int_{-T}^T dt H_I(t) \right] | 0 \rangle} \end{aligned}$$

5.0.4 Wick's Theorem and Contractions (real)

- A contraction in a Feynman diagram is when

$$\begin{aligned} &= 9D_{12}D_{xx}D_{xy}P_{yy} + 6D_{12}D_{xy}^3 \\ &+ 18D_{1x}D_{2x}D_{xz}P_{yy} + 9D_{12}D_{xy}D_{xx}P_{yy} + \\ &18D_{12}D_{xy}D_{xy}^2 + 18D_{12}D_{xy}P_{yy}D_{xy}D_{xx} \\ &+ 9D_{1y}D_{xx}D_{xx}D_{yy} + 18D_{1y}D_{2x}D_{xy}^2 \end{aligned}$$

6 Spinors

6.1 Lorentz and Spinors

Some Supplementary material for this section is Spinors for Beginners by Eigenchris which is very useful to get a grasp of the actual math that is going on.

6.1.1 Lorentz Transformations

Definition 6.1.1.1 (Metric Tensor). A (or in special relativity, *The*) metric tensor is defined as being the dot product of basis vectors

$$g_{\mu\nu} = \hat{e}_{(\mu)} \cdot \hat{e}_{(\nu)}$$

This definition is all but useless in Quantum Field Theory due to the whole theory entirely using Special Relativity instead of General Relativity¹

The special relativity metric tensor (which is subject to arguments over convention), is either $g_{\mu\nu} = \text{diag}(+, -, -, -)$ or $g_{\mu\nu} = \text{diag}(-, +, +, +)$ ².

Definition 6.1.1.2 (Lorentz Transformation & Lorentz Invariance). A *Lorentz transformation* Λ is defined as a coordinate transformation

¹Unless string theory, again

²Personally I am used to $\text{diag}(+, -, -, -)$.

$$\Lambda : (t, x, y, z) \rightarrow (t', x', y', z')$$

such that

$$\Lambda^T g \Lambda = g$$

or

$$\Lambda^\mu_\alpha \Lambda^\nu_\beta \eta_{\mu\nu} = \eta_{\alpha\beta}$$

In special relativity (minkowski space), this condition is loosened to

$$s^2 = t^2 - x^2$$

is conserved but the previous condition is still useful in discussion of group theory and derivation of *Spinors*. Generally, Lorentz transformations are transformations that

6.1.2 Group Theory (cont'd)

Definition 6.1.2.1 (Lorentz Group). This is the set of operations Λ such that

$$\Lambda^T g \Lambda = g$$

- For instance, a Lorentz transformation could follow the 4-vector representation, defined as

$$x_\mu \rightarrow \Lambda_{\mu\nu} x_\nu$$

- More generally, a field ϕ transforms under a representation of the lorentz group as

$$\phi_{i(x)} \rightarrow R(\Lambda)_{ij} \phi_j(\Lambda^{-1}x)$$

(Similar to a matrix transformation)

- The continuous Lorentz transformations that exist are *rotations* and *boosts*, which are the most general possible (continuous) Lorentz transformations which can occur as they (+ the discrete Lorentz transformations such as P and T) form the **Lorentz Group**
- Rotations can be represented, again in the 4-vector representation, as rotation matrices. We let $\theta_x, \theta_y, \theta_z$ represent rotations in the x, y, z axes respectively. Then, the rotation matrices can be represent as as

$$\Lambda(\theta_x) = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & \cos \theta_x & \sin \theta_x \\ & & -\sin \theta_x & \cos \theta_x \end{pmatrix}, \Lambda(\theta_y) = \begin{pmatrix} 1 & & & \\ & \cos \theta_y & -\sin \theta_y & \\ & & 1 & \\ & \sin \theta_y & & \cos \theta_y \end{pmatrix}$$

$$\Lambda(\theta_z) = \begin{pmatrix} 1 & & & \\ & \cos \theta_z & \sin \theta_z & \\ & -\sin \theta_z & \cos \theta_z & \\ & & & 1 \end{pmatrix}$$

- In a similar manner, boosts can be represented by matrices in the 4-vector representation. Denoting $\beta_x, \beta_y, \beta_z$ as the boost angles (in the hyperbolic trigonometric functions), we have that

$$\Lambda(\beta_x) = \begin{pmatrix} \cosh \beta_x & \sinh \beta_x & & \\ \sinh \beta_x & \cosh \beta_x & & \\ & & 1 & \\ & & & 1 \end{pmatrix}, \Lambda(\beta_y) = \begin{pmatrix} \cosh \beta_y & \sinh \beta_y & & \\ & 1 & & \\ \sinh \beta_y & & \cosh \beta_y & \\ & & & 1 \end{pmatrix}$$

$$\Lambda(\beta_z) = \begin{pmatrix} \cosh \beta_z & & \sinh \beta_z & \\ & 1 & & \\ & & 1 & \\ \sinh \beta_z & & & \cosh \beta_z \end{pmatrix}$$

- Can utilize *infinitesimal* Lorentz transformations in order to extract the group ($O(1, 3)$) (we let β_i parametrize boosts and θ_i parametrize rotations)

►

$$\delta x_0 = \beta_i x_i$$

$$\delta x_i = \beta_i x_0 - \varepsilon_{ijk} \theta_j x_k$$

or

$$\delta x_\mu = i \left[\sum_{i \in \{1,2,3\}} \theta(J_i)_{\mu\nu} + \beta_i (K_i)_{\mu\nu} \right] x_\nu$$

- J_i and K_i matrices:

$$J_1 = i \begin{pmatrix} 0 & & & \\ & 0 & & \\ & & 0 & -1 \\ & & 1 & 0 \end{pmatrix}, J_2 = i \begin{pmatrix} 0 & & & \\ & 0 & 1 & \\ & & 0 & \\ -1 & & & 0 \end{pmatrix}, J_3 = i \begin{pmatrix} 0 & & & \\ & 0 & -1 & \\ & 1 & 0 & \\ & & & 0 \end{pmatrix}$$

$$K_1 = i \begin{pmatrix} 0 & -1 & & \\ -1 & 0 & & \\ & & 0 & \\ & & & 0 \end{pmatrix}, K_2 = i \begin{pmatrix} 0 & & -1 & \\ & 0 & & \\ -1 & & 0 & \\ & & & 0 \end{pmatrix}, K_3 = i \begin{pmatrix} 0 & & & -1 \\ & 0 & & \\ & & 0 & \\ -1 & & & 0 \end{pmatrix}$$

- ▶ Such matrices are *generators of the Lorentz group*
 - Can use to obtain the Lie Algebra
- ▶ $\Lambda = \exp(i\theta_i J_i + i\beta_i K_i)$

6.2 Dirac Spinors (TODO)

6.2.1 Spinors

Definition 6.2.1.1 (Dirac Spinor). An object ψ obeying the transformation properties

$$\psi \rightarrow S[\Lambda]\psi, \psi^\dagger \rightarrow \psi^\dagger S[\Lambda]$$

- What does this mean?
- A lorentz transformation can be written as

$$\Lambda_\nu^\mu = \exp\left(\frac{1}{2}\Omega_{\rho\sigma}(J^{\rho\sigma})_\nu^\mu\right)$$

with new notation introduced for convenience such that $\Omega_{0i} = \beta_i$ boosts, $\Omega_{12} = -\Omega_{21} = \theta_z$ (rotation angles) with $(J^{\rho\sigma})^{\mu\nu} = g^{\rho\mu}g^{\sigma\nu} - g^{\sigma\mu}g^{\rho\nu}$

- ▶ These are another way to write generators for the lorentz group
- ▶ Notice that, using this tensor, we can write

$$K_i = -i(J^{0i})_\nu^\mu$$

for boosts and

$$J_i = \frac{i}{2}\varepsilon_{ijk}(J^{jk})_\nu^\mu$$

for rotations

- These generators form $\mathfrak{so}(1, 3)$, the Lorentz Algebra
 - ▶ Their exponential gives a representation of the Lorentz Group double cover $\text{Spin}(1, 3) \cong \text{SL}(2, \mathbb{C})$

Theorem 6.2.1.2. *The Lorentz Algebra, $\mathfrak{so}(1,3)$, has two commuting sub-algebras represented by the 3d rotation algebras $\mathfrak{su}(2)$*

Proof.

- Notice that if we switch back to J_i and K_i , we can derive $SU(2)$, the group for spinors.
- Define $J_i^+ \equiv \frac{1}{2}(J_i + iK_i)$, $J_i^- \equiv \frac{1}{2}(J_i - iK_i)$
- We have then,

$$\begin{aligned} [J_i^+, J_j^+] &= i\varepsilon_{ijk} J_k^+ \\ [J_i^-, J_j^-] &= i\varepsilon_{ijk} J_k^- \\ [J_i^+, J_j^-] &= 0 \end{aligned}$$

hence, $\mathfrak{so}(1,3)$ has 2 commuting sub algebras, both of which are 3d rotation groups given by the Levi-Cevita connections. Hence, $SO(1,3) \cong SU(2) \oplus SU(2)$

□

- Hence, we now have the definition of a spinor

Definition 6.2.1.3 (Spinor). A *Spinor* is an object that transforms under $SU(2)$

6.2.2 Clifford Algebra

- To construct Dirac Spinors, one must first construct gamma matrices, which form the basis of a Clifford Algebra $\mathcal{C}\ell_{1,3}(\mathbb{R})$

Definition 6.2.2.1 (Gamma Matrices). Gamma matrices are defined as being anticommuting matrices such that

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}$$

and having the property that

$$(\gamma^0)^2 = 1, (\gamma^i)^2 = -1$$

- As seen in the *math toolkit*, these matrices can be represented as

$$\gamma^\mu = \begin{pmatrix} \sigma^\mu & \\ \bar{\sigma}^\mu & \end{pmatrix}$$

with $\sigma^\mu = (\mathbb{1}, \vec{\sigma})$, $\bar{\sigma}^\mu = (\mathbb{1}, -\vec{\sigma})$ and σ^i being Pauli Matrices

- This representation is known as the Weyl Representation
- An additional piece of useful notation is the idea that $\not{p} = p_\mu \gamma^\mu = p^\mu \gamma_\mu$

6.2.3 Pauli Matrix and Gamma Matrix Identities

$$\sigma^i \sigma^j = \delta^{ij} i \epsilon^{ijk} \sigma^k, \{\sigma^i, \sigma^j\} = 2\delta^{ij}, [\sigma^i, \sigma^j] = 2i\epsilon^{ijk} \sigma^k$$

For Pauli Matrices

- A useful quantity to define is a representation of the Lorentz Algebra

$$S^{\rho\sigma} = \frac{1}{2}[\gamma^\rho, \gamma^\sigma] = \frac{1}{2}\gamma^\rho \gamma^\sigma - \frac{1}{2}\gamma^\sigma \gamma^\rho$$

6.2.4 Lorentz Group Representation

- We now want to derive why spinors behave the way that they do. That is, why do they have spin $\frac{1}{2}$?
- As seen before, we have our generators written as linear combinations

$$J_i^+ \equiv \frac{1}{2}(J_i + iK_i), \quad J_i^- \equiv \frac{1}{2}(J_i - iK_i)$$

which formed two commuting subgroups $SO(1, 3) \cong SU(2) \times SU(2)$

- There are rotation generators $\vec{J}$ and hence the Lorentz group is a super of $SO(3)$
 - Any representation of $SO(3)$ is a representation of the Lorentz Group
- Notice that $SU(2)$ both have quantum numbers A and B . The spin is their sum.

$\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ representation	(A, B)	$(0, 0)$	$(\frac{1}{2}, 0)$	$(0, \frac{1}{2})$	$(\frac{1}{2}, \frac{1}{2})$	$(1, 1)$
$\mathfrak{so}(3)$ representation	$j = A + B$	0	$\frac{1}{2}$	$\frac{1}{2}$	$1 \oplus 0$	$2 \oplus 1 \oplus 0$

- In this table, you can see that there are two different representations of $j = \frac{1}{2}$. Therefore, there are two different type of spinors (this corresponds to chirality which you will see)
- Now we 2×2 matrices such that

$$[J_i^+, J_j^+] = i\epsilon_{ijk} J_k^+$$

$$[J_i^-, J_j^-] = i\epsilon_{ijk} J_k^-$$

$$[J_i^+, J_j^-] = 0$$

- Remember our sigma matrix identities?, we had that

$$\left[\frac{\sigma_i}{2}, \frac{\sigma_j}{2} \right] = i\varepsilon_{ijk} \frac{\sigma_k}{2}$$

which is the perfect $\mathfrak{so}(3)$ algebra that we wanted

- For the $(\frac{1}{2}, 0)$ representation, we can set $J^+ = 0$ and $J_i^- = \frac{1}{2}\sigma_i$ as our generators
- For the $(0, \frac{1}{2})$ representation, we can set $J^- = 0$ and $J_i^+ = \frac{1}{2}\sigma_i$ as our generators
- Now we are able to get our Lorentz transformations.
 - For rotations $\vec{J} = \vec{J}^- + \vec{J}^+$
 - For our boosts, $\vec{K} = i(\vec{J}^- - \vec{J}^+)$
- Thus,

$$\begin{aligned} \left(\frac{1}{2}, 0 \right) : \vec{J} &= \frac{1}{2}\vec{\sigma}, \quad \vec{K} = \frac{i}{2}\vec{\sigma} \\ \left(0, \frac{1}{2} \right) : \vec{J} &= \frac{1}{2}\vec{\sigma}, \quad \vec{K} = -\frac{i}{2}\vec{\sigma} \end{aligned}$$

6.2.5 Note: Hermiticity

- Rotations are hermitian and boosts are anti hermitian
- Now we have an even better way to define spinors in terms of their handedness

Definition 6.2.5.1 (Right Handed Weyl Spinor).

- An object transforming under the $(0, \frac{1}{2})$ representation of the Lorentz group
- Right handed weyl spinors transform as

$$\psi_R \rightarrow e^{\frac{1}{2}(i\theta_j\sigma_j + \beta_j\sigma_j)}\psi_R = (1 + \frac{i}{2}\theta_j\sigma_j + \frac{1}{2}\beta_j\sigma_j + \dots)\psi_R$$

Definition 6.2.5.2 (Left Handed Weyl Spinor).

- An object transforming under the $(\frac{1}{2}, 0)$ representation of the Lorentz group
- These transform as

$$\psi_L \rightarrow e^{\frac{1}{2}(i\theta_j\sigma_j - \beta_j\sigma_j)}\psi_L = (1 + \frac{i}{2}\theta_j\sigma_j - \frac{1}{2}\beta_j\sigma_j + \dots)\psi_L$$

- There are also infinitesimal transformations which are

$$\delta\psi_R = \frac{1}{2}(i\theta_j + \beta_j)\sigma_j\psi_R$$

$$\delta\psi_L = \frac{1}{2}(i\theta_j - \beta_j)\sigma_j\psi_L$$

$$\delta\psi_R^\dagger = \frac{1}{2}(-i\theta_j + \beta_j)\psi_R^\dagger\sigma_j$$

$$\delta\psi_L^\dagger = \frac{1}{2}(-i\theta_j - \beta_j)\psi_L^\dagger\sigma_j$$

6.2.6 note: Unitarity

Since there is no finite dimensional unitary representation of the lorentz group, spinor bases must be built in terms of momentum, much like the photon field

6.2.7 Lorentz Invariant Lagrangian

- How do we get a Lorentz invariant lagrangian with our new spinor valued fields?
- First, let us write
 - $(\frac{1}{2}, 0)$ rep:

$$\psi_R(x) = \begin{pmatrix} \psi_{1R}(x) \\ \psi_{2R}(x) \end{pmatrix}$$

- $(0, \frac{1}{2})$ rep:

$$\psi_L(x) = \begin{pmatrix} \psi_{1L}(x) \\ \psi_{2L}(x) \end{pmatrix}$$

- We want a lagrangian with the right number of DOF, of course
- Let us be naive and write

$$\mathcal{L} = (\psi_R)^\dagger \square \psi_R + m^2 (\psi_R)^\dagger \psi_R$$

- not L.I.:

$$\begin{aligned} \delta(\psi_R^\dagger \psi_R) &= \frac{1}{2}\psi_R^\dagger [i\theta_j + \beta_j\sigma_j]\psi_R + \frac{1}{2}[\psi_R^\dagger (-i\theta_j + \beta_j)\sigma_j]\psi_R \\ &= \beta_i\psi_R^\dagger \sigma_i \psi_R \neq 0 \end{aligned}$$

- Therefore, we must have two fields ψ_R and ψ_L
 - Let's try terms like $\psi_L^\dagger \psi_R$:

$$\delta(\psi_L^\dagger \psi_R) = \left[\psi_L^\dagger \frac{i}{2}(-i\theta_i - \beta_i)\sigma_i \right] \psi_R + \psi_L^\dagger \left[\frac{i}{2}(i\theta_i + \beta_i)\sigma_i \psi_R \right] = 0$$

- Adding the h.c. will give us a lagrangian:

$$\mathcal{L}_{\text{Dirac Mass}} = m(\psi_L^\dagger \psi_R + \psi_R^\dagger \psi_L)$$

- Now we want kinetic terms to add to the lagrangian
 - we can try adding terms like $\mathcal{L}_{\text{kin}} = \psi_L^\dagger \square \psi_R + \psi_R^\dagger \square \psi_L$ but these are “not interesting”
 - Let's look at a term like $\psi_R^\dagger \sigma_i \psi_R$: [

$$\begin{aligned} \delta(\psi_R^\dagger \sigma_i \psi_R) &= \frac{1}{2} \psi_R^\dagger \sigma_i [(i\theta_j + \beta_j) \sigma_j \psi_R] + \frac{1}{2} [\psi_R^\dagger (-i\theta_j + \beta_j) \sigma_j] \sigma_i \psi_R \\ &= \frac{\beta_j}{2} \psi_R^\dagger (\sigma_i \sigma_j + \sigma_j \sigma_i) \psi_R + \frac{i\theta_j}{2} (\sigma_i \sigma_j - \sigma_j \sigma_i) \psi_R \\ &= \beta_i \psi_R^\dagger \psi_R - \theta_j \epsilon_{ijk} \psi_R^\dagger \sigma_k \psi_R \end{aligned}$$

]

$$\therefore \delta(\psi_R^\dagger \psi_R, \psi_R^\dagger \sigma_i \psi_R) = (\beta_i \psi_R^\dagger \sigma_i \psi_R, \beta_i \psi_R^\dagger \psi_R - \epsilon_{ijk} \theta_j \psi_R^\dagger \sigma_k \psi_R)$$

- Notice that this is exactly how a vector transforms
 - $\delta(v_0, v_i) = (\beta_i v_i, \beta_i v_0 - \epsilon_{ijk} \theta_j v_k)$

7 Quantum Electrodynamics

Definition 7.1 (Quantum Electrodynamics). Quantum Electrodynamics is a relatively simple theory with only one force involved (as suggested by the name, it is the electromagnetic force), mediated by the *photon* A_μ and one/two type(s) of fermion depending on what you consider a type of particle (an *electron/positron*)

7.0.1 Charge

- The charge in quantum electrodynamics is associated with the $U(1)$ symmetry

$$\phi \rightarrow e^{-i\alpha(x)} \phi$$

(in fact, this applies in spinor QED as well, due to the)

•

7.1 Spinor QED (Standard Quantum Electrodynamics)

- The Spinors in Quantum Electrodynamics are Dirac Spinors due to their presence of charge (Since Majorana spinors are the same as their antiparticle, they of course cannot be a candidate for this, and hence the spinors must be Dirac Spinors).

7.1.1 Solution to the Dirac Equation

- We want to quantize our dirac spinors ψ .
- We follow a similar procedure to what we did with photons (we will introduce basis spinors similar to polarizations ε_μ)
- The dirac equation

$$(i\not{\partial} - m)\psi = 0$$

implies that, for spinors

$$\left((i\partial_\mu - eA_\mu)^2 - \frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu} - m^2 \right)\psi = 0$$

- For antispinors,

$$\bar{\psi} \left((i\tilde{\partial}_\mu + eA_\mu)^2 + \frac{e}{2}F_{\mu\nu}\sigma^{\mu\nu} - m^2 \right) = 0$$

- Similarly to a Scalar field theory, we will quantize particles by writing

$$\psi_s(x) = \int \frac{d^3p}{(2\pi)^3} u_s(p) e^{ipx}$$

for particles and

$$\chi_s(x) = \int \frac{d^3p}{(2\pi)^3} v_s(p) e^{ipx}$$

for antiparticles

- These $u_s(p)$ and $v_s(p)$ are *basis spinors* which satisfy the momentum space dirac equation. That is,

$$\begin{pmatrix} -m & p \cdot \sigma \\ p \cdot \bar{\sigma} & -m \end{pmatrix} u_s(p) = \begin{pmatrix} -m & -p \cdot \sigma \\ -p \cdot \bar{\sigma} & -m \end{pmatrix} v_s(p) = 0$$

- Again, these $u_s(p)$ and $v_s(p)$ are 4-index Dirac spinors.

7.1.2 Basis Dirac Spinors

In general, the solution to this equation is

$$u_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \xi \\ \sqrt{p \cdot \bar{\sigma}} \xi \end{pmatrix}, v_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta_s \\ -\sqrt{p \cdot \bar{\sigma}} \eta_s \end{pmatrix}$$

- $$\sum_s u_s(p) \bar{u}_s(p) = \not{p} + m$$
- $$\sum_s v_s(p) \bar{v}_s(p) = \not{p} - m$$

7.1.3 CPT Theorem

- In order to have the entire Lorentz Group, we must define the following discrete transforms for spinors.
- Time reversal T which maps

$$T : (t, \vec{x}) \rightarrow (-t, \vec{x})$$

- Parity P which maps

$$P : (t, \vec{x}) \rightarrow (t, -\vec{x})$$

- Charge conjugation

$$C : \psi \rightarrow -i\gamma^2\bar{\psi}$$

reverses charge (essentially it is particle $\rightarrow$ antiparticle)

7.1.4 Interacting QED

- The Lagrangian for Quantum Electrodynamics is

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\bar{\psi}\not{D}\psi - m\bar{\psi}\psi \\ &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi - e\bar{\psi}\gamma^\mu\psi A_\mu\end{aligned}$$

- This gives rise to the interaction $-ie\gamma^\mu$ with the Feynman Diagram

Theorem 7.1.4.1 (SPIN-STATISTICS THEOREM). *The Spin statistics theorem is the theorem that spinors are governed by Fermi-Dirac statistics and that Vectors are governed by Bose-Einstein statistics. Essentially $\psi(x)\chi(y) = -\chi(x)\psi(y)$ and*

- The Feynman propagator for spinors is

$$\langle 0|T\psi(0)\bar{\psi}(x)|0\rangle = \int \frac{d^4p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} e^{ipx}$$

with of course the implicit limit of $\varepsilon \rightarrow 0$

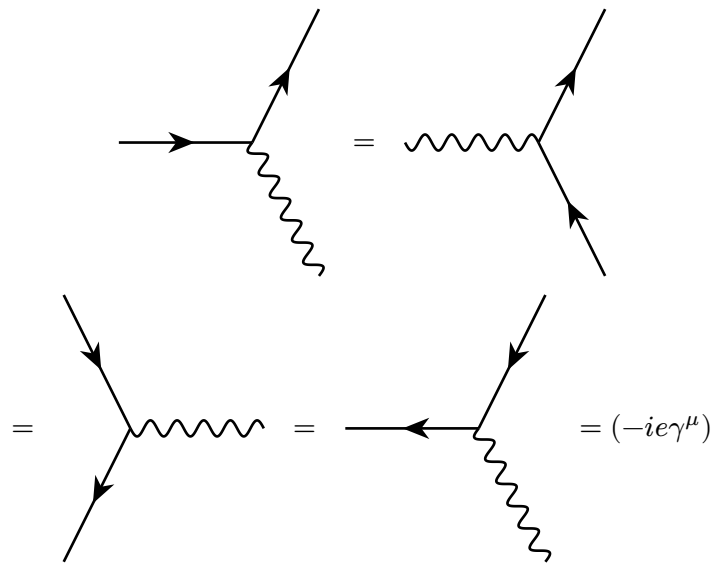
- The derivation for this is as follows

►

- The Feynman propagator clearly gives rise to the electron propagator present on internal electrons

$$\text{feynman diagram} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon}$$

- The QED vertex factor is given by the interaction term $-e\bar{\psi}\gamma^\mu A_\mu\psi$



- Notice how all of these terms are equivalent (as opposed to scalar QED)
 - This is one of the many differences between spinor QED and scalar QED
- Another difference in Spinor QED is the fact that it has no seagull interaction allowed by an $A^2\phi^2$ term in scalar QED, simply because this term doesn't exist in QED

8 Renormalization

“The shell game that we play ... is technically called ‘renormalization’. But no matter how clever the word, it is still what I would call a dippy process! Having to resort to such hocus-pocus has prevented us from proving that the theory of quantum electrodynamics is mathematically self-consistent. It’s surprising that the theory still hasn’t been proved self-consistent one way or the other by now; I suspect that renormalization is not mathematically legitimate.”

— Richard Feynman

8.1 Stereotypical Examples of (Counterterm) Renormalization

- Renormalization is the process of removing infinities by putting them in “bare” variables
- The three most stereotypical examples of renormalization of quantum electrodynamics (and *in general*) are the following
 1. Vacuum Polarization

- 2. Electron Self Interaction
- 3. The anomalous magnetic moment
- There are two types of renormalization:
 1. Counterterm renormalization
 2. Wilsonian renormalization
 - Arguably the coolest type of renormalization but it seems like it's completely different than traditional counterterm renormalization

8.1.1 Mass Renormalization

- This is equivalent to the Electron self interaction

Why Renormalize?

- *In general, divergences form due to loop diagrams which give infinite integrals*

Definition 8.1.1.1 (Superficial Degree of Divergence).

- This quantity is a very standard quantity which counts essentially how badly something diverges.
 - In QED, define the following quantities :

$$\begin{aligned}
 N_e &= \# \text{ elec} \\
 N_\gamma &= \# \text{ phot} \\
 P_e &= \# \text{ electron prop.} \\
 P_\gamma &= \# \text{ photon prop.} \\
 V &= \# \text{ vertices} \\
 L &= \# \text{ loops} \\
 D &= 4L - P_e - 2P_\gamma
 \end{aligned}$$

- The degree of divergence is $= 2(P_e + P_\gamma - V + 1) - P_e - 2P_\gamma$
- Now we will make sense of this quantity

Definition 8.1.1.2 (PV Regulation).

- Consider a divergent integral, such as

$$\int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2}$$

- The superficial degree of divergence is the difference in powers of k in the numerator and denominator.

- That is, $D_{\text{ex.}} = 2$.
- PV regulation works by giving the integral a cutoff

$$\int^\Lambda \frac{d^4k}{(2\pi)^4} \frac{1}{k^2} \propto \Lambda^2$$

- We would then enter counterterms into the lagrangian which would precisely (or not so precisely in some schemes like $\overline{\text{MS}}$ as you will see) cancel them out
- Now we see that, in fact, the degree of divergence counts the power of Λ . If $D = 0$, it is proportional to $\ln \Lambda$
- There is also another way to evaluate the integrals which shows us a divergence

Definition 8.1.1.3 (DimReg).

- DimReg (dimension regulation) is a method of scanning for divergences by setting $d = 4 - \varepsilon$ and taking $\lim_{\varepsilon \rightarrow 0}$ of the integral
 - You only take this limit after you subtract the divergence
- Lets look at the actual electron self energy to see what we can analyze
- At 1 loop,

$$\langle \Omega | T \bar{\psi}(x) \psi(x) | \Omega \rangle = \text{feynman diagrams} = \text{normal electron propagator} \\ + \text{momentum integral of some function } f$$

- We define

$$f \equiv \frac{i(\not{p} + m)}{p^2 - m_0^2 + i0} (-\Sigma_2(\not{p})) \frac{i(\not{p} + m_0)}{p^2 - m_0^2 + i0}$$

- Essentially, we want to evaluate the loop integral given by

$$i\Sigma_2(\not{p}) = (-ie)^2 \int \frac{d^4 k}{(2\pi)^4} \gamma^\mu \frac{i(\not{k} + m_0)}{k^2 - m_0^2 + i0} \gamma^\nu \frac{-ig_{\mu\nu}}{(p - k)^2 + i0}$$

- This trick that we use to evaluate these integrals is called *feynman parameterization*
- Notice that

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{(xA + (1-x)B)^2}$$

- Taking $A = (p - m_0)^2 - m_0^2 + i0$ and $B = k^2 - m_0^2 + i0$, we obtain that

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{[x^2 - 2xk \cdot p + xp^2 - x\mu^2 - (1-x)m_0^2 + i\epsilon + x^2p^2 - x^2p^2]^2}$$

8.1.2 Vacuum Polarization

- This is the photon splitting into an electron and a positron and then them combining again into a photon
- The vertex evaluates to

$$\text{feyn diag} = -(-ie)^2 \int \frac{d^4 k}{(2\pi)^4} \frac{i}{(p-k)^2 - m^2} \frac{i}{k^2 - m^2} \\ \times \text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)]$$

$$\text{tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)] = 4[-p^\mu k^\nu - k^\mu p^\nu + 2k^\mu k^\nu + g^{\mu\nu}(-k^2 + p \cdot k + m^2)]$$

- We define the loop in this diagram as

$$i\Pi_2^{\mu\nu} \equiv \int \frac{d^4 k}{(2\pi)^4} \frac{i}{(p-k)^2 - m^2} \frac{i}{k^2 - m^2} \times \text{Tr}[\gamma^\mu (\not{k} - \not{p} + m) \gamma^\nu (\not{k} + m)]$$

- With our trace, we get

$$i\Pi_2^{\mu\nu} = -4e^2 \int d^4 k \frac{2k^\mu k^\nu + g^{\mu\nu}(-k^2 + p \cdot k + m^2)}{[(p-k)^2 - m^2 + i\epsilon][k^2 - m^2 + i\epsilon]}$$

- We can parametrize this using a trick called feynman parameterization where we take $k^\mu \rightarrow k^\mu + p^\mu(1-x)$ and integrate

$$\Pi_2^{\mu\nu} = 4ie^2 \int d^4 k \int_0^1 dx \frac{2k^\mu k^\nu - g^{\mu\nu}[k^2 - x(1-x)p^2 - m^2]}{[k^2 + p^2 x(1-x) - m^2]^2} \\ = -8p^2 g^{\mu\nu} \frac{e^2}{(4\pi)^{d/2}} \Gamma(2-d/2) \mu^{4-d} \\ \times \int_0^1 dx x(1-x) \left(\frac{1}{m^2 - p^2 x(1-x)} \right)^{2-d/2}$$

- This yields

$$\Pi_2^{\mu\nu} = -\frac{e^2}{2\pi^2} p^2 g^{\mu\nu} \int_0^1 dx x(1-x) \left[\frac{2}{\epsilon} + \ln \left(\frac{\tilde{\mu}^2}{m^2 - p^2 x(1-x)} \right) + O(\epsilon) \right]$$

8.2 Wilsonian Renormalization

- This is such an important topic in quantum field theory that I considered giving it its own section
 - Wilsonian renormalization has application in not only quantum field theory, but effectively all of physics in the language of condensed matter physics which is why it is an amazing feat that should not be overlooked

- Wilsonian renormalization tells us how we can neglect extremely high energy terms which decay exponentially with a high cutoff (as you will see)

8.2.1 Running Coupling

- We must first define the *running coupling* of a theory before discussing wilsonian renormalization
- The running coupling is defined through the β -function

Definition 8.2.1.1 (Beta Function). The beta function is defined as

$$\beta(g) \equiv \mu \frac{d}{d\mu} g$$

- This represents how the renormalized coupling constants of a theory “run” with the scale at which the theory is measured.
- The β function of a theory is equivalent to

$$\beta(g) = \Lambda \frac{d}{d\Lambda} g$$

where the Λ is interpreted as the energy scale which will be shown.

•

9 Quantum Chromodynamics

9.1 Non Abelian Gauge Theories

- Non Abelian gauge theories are the fundamentals of QCD (in fact, they ARE Quantum Chromodynamics).
 - We need to establish quite a bit more formal background in order to do actual physics, otherwise nothing would make sense

9.1.1 Lie Groups and Lie Algebras

Definition 9.1.1.1 (Non Abelian Gauge Group).

- Suppose that you have a representation of a Lie Group, written as

$$R(G) = U = \exp(i\theta^a T^a)$$

- θ^a is a number
- T^a are the group generators for the *Lie Algebra* that generates the *Lie Group* G .
- The generators T^a satisfy the commutation

$$[T^a, T^b] = i f^{abc} T^c$$

- If $f^{abc} = 0$, the gauge group G is abelian

- If $f^{abc} \neq 0$, the gauge group G is *non abelian*
- The commutation relations [...] are a mapping $\mathcal{G} \times \mathcal{G} \rightarrow \mathcal{G}$, without having to have a well-defined product
 - In the special case that a product is well defined, $[A, B] = AB - BA$
- Note: The **Jacobi Identity** must be satisfied regardless of the product being well defined

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0$$

- By the Jacobi Identity,

$$f^{abd} f^{dce} + f^{bcd} f^{dae} + f^{cad} f^{dbe} = 0$$

Definition 9.1.1.2 (Ideal).

- An *ideal* subalgebra $\mathfrak{I} \subset \mathfrak{g}$ is one that satisfies $[s, i] \in \mathfrak{I} \forall g \in \mathfrak{g}, i \in \mathfrak{I}$

Definition 9.1.1.3 (Simple).

- A simple Lie Algebra $\mathfrak{g}$ is one that has no trivial *ideal* subalgebras

Example 9.1.1.4 (Simple Lie Algebras).

- Very common Simple Lie Groups/Algebras are $\mathfrak{so}(N)$ and $\mathfrak{su}(N)$
- There is also the symplectic group $\mathfrak{sp}(N)$
- All of these you will soon see

Definition 9.1.1.5 (Semisimple).

- A semisimple Lie algebra $\mathfrak{g}$ is one that can be expressed as the direct sum of simple Lie Algebras $\mathfrak{g}_1, \mathfrak{g}_2, \dots, \mathfrak{g}_n$
 - That is,

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3 \oplus \dots \oplus \mathfrak{g}_n$$

- The standard model is a very common example of a Semisimple Lie Group
 - The Lie group for the Standard model is

$$G_{\text{SM}} = \text{SU}(3) \otimes \text{SU}(2) \otimes U(1)$$

- The Lie algebra for the Standard model is

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$$

Theorem 9.1.1.6 (HERMICITY).

- All finite dimensional representations of semisimple Lie algebras are hermitian

Proof Hermiticity.

- Unfortunately I have tried and failed to prove it, so at this point i cannot provide a proof

□

Definition 9.1.1.7 (Unitary Groups).

- A unitary group is a group whose representation satisfies

$$\langle \vec{\psi} | \vec{\chi} \rangle = \langle \vec{\psi} | U^\dagger U | \vec{\chi} \rangle$$

- i.e. the inner product is conserved
- This directly implies that $U^\dagger U = \mathbb{1}$
- Elements of $SU(N)$ have determinant 1
 - In general, matrix representations of unitary groups always have determinant 1 due to their being unitary
- The group $SU(N)$ is defined by its action upon N dimensional vector spaces

$$U = \exp(i\theta^a T^a)$$

- In $SU(N)$, T^a are always hermitian
- There are $N^2 - 1$ generators for $SU(N)$
 - This implies that the dimension $d(G)$ is equal to $d(G) = N^2 - 1$

Definition 9.1.1.8 (Orthogonal Group).

- Orthogonal groups preserve a strictly real inner product
- $V \cdot W = V \cdot O^T \cdot O \cdot W$
- This directly implies that $O^T O = \mathbb{1}$
- The dimension of an orthogonal group $SO(N)$ is $d(SO(N)) = \frac{1}{2}d(SU(N)) = \frac{1}{2}N(N-1)$
 - This is due to the fact that there are half as many generators for orthogonal groups due to their strict realness
- All elements of an orthogonal group (not special orthog.) have determinant $\det(O) = \pm 1$
 - In $SO(N)$, the matrices are the ones with $\det(O) = +1$
 - This has no impact on the dimension of the group

Definition 9.1.1.9 (Symplectic Group).

- Symplectic groups $Sp(N)$ are defined via preserving a quaternionic inner product
- They satisfy

$$\Omega S = -S^T \Omega$$

where $S \in Sp(N)$ and

$$\Omega = \begin{pmatrix} 0 & \mathbb{I} \\ -\mathbb{I} & 0 \end{pmatrix}$$

- There are, finally, the *exceptional* simple groups G_2, F_4, E_6, E_7, E_8 which are of no use to standard model particle physics
- The *Only* finite dimensional simple Lie Algebras are the algebras for the exceptional simple groups and $\mathfrak{so}(N), \mathfrak{su}(N), \mathfrak{sp}(N)$

9.1.2 Representations

- Consider representations of $SU(N)$ groups
- Free theories of N fields are invariant under $U(1) \times SU(N)$
 - Since $SU(N)$ is simply connected, any representation of $SU(N)$ is one to one representation of $\exp(\mathfrak{su}(N))$

Example 9.1.2.1 (Two Fields).

- Consider 2 fields ϕ_1, ϕ_2 with a kinetic lagrangian

$$\mathcal{L}_{kin} = (\partial_\mu \phi_1^*)(\partial_\mu \phi_1) + (\partial_\mu \phi_2^*)(\partial_\mu \phi_2) = (\partial_\mu \vec{\phi})^\dagger (\partial_\mu \vec{\phi})$$

- We can write

$$\vec{\phi} = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix}$$

- This is invariant under a global $SU(2)$ symmetry $\vec{\phi} \rightarrow U \vec{\phi}$
- Generally, as we have seen, we can write

$$U = \exp[i(\alpha_1 \tau_1 + \alpha_2 \tau_2 + \alpha_3 \tau_3)] = \exp(ia^a \tau^a)$$

$$w/\tau^a = 1/2\sigma^a$$

- Such τ^a satisfy

$$\tau^a : [\tau^a, \tau^b] = i\epsilon^{abc} \tau^c$$

- This provides $f^{abc} = \epsilon^{abc}$

Definition 9.1.2.2 (Fundamental).

- A fundamental representation is the smallest non trivial representation of a Lie Algebra
- For $SU(N)$, the fundamental representation is the set of $N \times N$ hermitian matricians with determinant 1
- Under the fundamental representation, a set of N fields transform (infinitesimally) as

$$\phi_i \rightarrow \phi_i + i\alpha^a (T_{\text{fund}}^a)_{ij} \phi_j$$

for some $\alpha^a \in \mathbb{R}$

Definition 9.1.2.3 (Anti-fundamental Representation).

- $T_{\text{anti-fund}}^a = -(T_{\text{fund}}^a)^*$
- The anti-fund representation of a group is the conjugate of the fundamental representation
- Notice that there is a minus sign in order to preserve the law $[T^a, T^b] = i f^{abc} T^c$
- Therefore, the infinitesimal transformation law for anti-fundamental is

$$\phi_i^* \rightarrow \phi_i^* + i\alpha^a (T_{\text{anti-fund}}^a)_{ij} \phi_j^* = \phi_i^* - i\alpha^a \phi_j^* (T_{\text{fund}}^a)_{ji}$$

- In order to determine the generators for a theory, we expand around $\mathbb{1}$
 - In SU(2), the generators are σ^μ

$$T^a = \tau^a \equiv \frac{\sigma^a}{2}$$

- In SU(3), the generators are *eight* (not $3 \otimes \bar{3} = 9$) **Gellmann matrices** (such that $T^a = \frac{\lambda^a}{2}$) which are as follows:

$$\lambda_1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda_2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\lambda_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \lambda_4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

$$\lambda_5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix} \quad \lambda_6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$\lambda_7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix} \quad \lambda_8 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$$

- Normalization of f^{abc} is arbitrary in QCD/non abelian gauge theory.
- A commonly used convention for normalization (in physics) is

$$\sum_{c,d} f^{acd} f^{bcd} = N \delta^{ab}$$

- In math, the convention is

$$\sum_{c,d} f^{acd} f^{bcd} = \delta^{ab}$$

- The fixing the normalization of structure constants implies that the normalization of generators is fixed *in all representations*
- For representations,

$$[T^a, T^b] = i f^{abc} T^c$$

must hold when f^{abc} is the same

- Notice that this equation is not invariant under a rescaling of T_R^c

- Normalization of $SU(N)$:
 - It is normalized s.t.

$$\text{tr}(T^a T^b) = \frac{1}{2} \delta^{ab}$$

- In a general lie algebra, $[T^a, T^b]$ is well defined and $T^a T^b$ is not (as seen before)
 - In the fundamental representation of $SU(N)$, such generators, in fact, can be multiplied

$$T^a T^b = \frac{1}{2N} \delta^{ab} + \frac{1}{2} d^{abc} T^c + \frac{1}{2} i f^{abc} T^c$$

$$d^{abc} = 2 \text{tr}[T^a \{T^b, T^c\}]$$

- In $SU(N)$, there is a unique d^{abc} up to a constant
 - In $SU(2)$, $d^{abc} = 0$
- Some useful trace identities for generators are the following:

$$\text{tr}[T^a T^b T^c] = \frac{1}{4} (d^{abc} + i f^{abc})$$

$$\text{tr}[T^a T^b T^c T^d] = \frac{1}{4N} \delta^{ab} + \frac{1}{8} (d^{abe} + i f^{abe}) \times (d^{cde} + i f^{cde})$$

Definition 9.1.2.4 (Adjoint Representation).

- The adjoint representation is the group acting upon its own elements
- Extremely formally

$$\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$$

with G an element of a Lie group, $\mathfrak{g}$ an element of a Lie algebra, and GL being the group of invertible matrices

- If the lie group generated by a lie algebra is e^X , then $\text{Ad}_A(X) = AXA^{-1}$ with $A \in G$.

- The adjoint representation amounts to acting upon the space that is spanned by the generators, as you have seen
- In $SU(N)$, there are $N^2 - 1$ generators and hence there are $N^2 - 1$ dimensions for the adjoint representation
- Matrices in the adjoint representation are given by $(T_{\text{adj}}^a)^{bc} = -if^{abc}$
 - For $SU(2)$, the matrices are

$$T_{adj}^1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, T_{adj}^2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, T_{adj}^3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- In $SU(3)$, they are 8×8 matrices as you may be able to guess
- *Gauge fields transform in the adjoint representation*

Definition 9.1.2.5 (Casimirs).

- Casimirs are used to “characterize representations” [1]

Definition 9.1.2.6 (Quadratic Casimir).

- The Quadratic Casimir is defined via

$$\left(\sum_a \right) T_R^a T_R^a = C_2(R) \mathbb{1}$$

Theorem 9.1.2.7 (SCHUR'S LEMMA).

- *Group elements that commute with all other elements are proportional to $\mathbb{1}$*
- This theorem is extremely nontrivial. You can see a proof in [2]
- I am, however, able to provide a proof on why the $T^a T^a$ commutes with everything

$$[T_R^a T_R^a, T_R^b] = (if^{abc} T_R^c) T_R^a + T_R^a (if^{abc} T_R^c) = -if^{abc} \{T_R^c, T_R^a\} = 0$$

- Evaluating quadratic Casimirs involve choosing generators such that

$$f_r [T_p^a, T_p^b] = T(R)^{ab}$$

- $T(R)$ is known as the *index*
- In the fundamental representation, the convention implies that
 - $T(\text{fund}) \equiv T_F = \frac{1}{2}$ since $(T^a)^i_j (T^a)^j_i = 1.2 \delta^i_i$
- In the adjoint representation, $T(\text{adj}) \equiv T_A = N$
 - This is due to the fact that $f^{acd} f^{bcd} = N \delta^{ab}$
- Setting $a = b$ and summing over a for the definition of $T(R)$ yields
 - $d(R) C_2(R) = T(R) d(G)$

- $d(R)$ is the dimension of the representation
- $d(G)$ is the dimension of the group

- Fundamental Casimir:

$$\triangleright C_F = C_2(fund) = \frac{N^2 - 1}{2N}$$

- In $SU(2)$, $C_F = \frac{3}{4}$
- In $SU(3)$, $C_F = \frac{4}{3}$

- Adjoint Casimir:

- $C_A \equiv C_2(adj) = N$
- Generally, terms in YM theories will contain at least some factors of C_F and C_A

- For any representation,

$$\text{tr}([T_R^a, T_R^b]T_R^c) = if^{abc} \text{tr}(T_R^d T_R^d) = if^{abc}T(R)$$

- This implies that

$$f^{abc} = -\frac{i}{T_F} \text{tr}([t^a, t^b]t^c)$$

- In $SU(N)$, the *Fierz Identity* is satisfied:

$$\triangleright \sum_a T_{ij}^a T_{kl}^a = \frac{1}{2} \left(\delta_{il} \delta_{jk} - \frac{1}{N} \delta_{ij} \delta_{kl} \right)$$

- Since generators in $SU(N)$ are traceless, the sum over δ^{ij} and δ^{kl} yields zero
- Hence,

$$\text{tr}[T^a A] \text{tr}[T^a B] = \frac{1}{2} [\text{tr}(AB) - \frac{1}{N} \text{tr}(A) \text{tr}(B)]$$

Definition 9.1.2.8 (Anomaly Coefficient).

- An anomaly coefficient $A(R)$ is defined by the equation

$$\text{tr}[T^a \{T^b, T^c\}] = \frac{1}{2} A(R) d^{abc} = A(R) \text{tr}[T^a \{T^b, T^c\}]$$

9.2 Wilson Lines

9.2.1 Abelian Wilson Lines

- The motivation between wilson lines is a want for telling if $\phi(x) = \phi(y)$ (some how)
- Notice that one cannot simply take the difference since

$$\phi(x) - \phi(y) \rightarrow e^{i\alpha(x)}\phi(x) - e^{i\alpha(y)}\phi(y)$$

- It is additionally impossible to take any notion of $\partial_\mu \phi$ due to derivative definition based on differences

Definition 9.2.1.1 (Wilson Line).

- A wilson line $W(x, y)$ is something that transforms as $W(x, y) \rightarrow e^{i\alpha(x)}W(x, y)e^{-i\alpha(y)}$

- This is a “bi-local transformation” [1]

- A key feature of Wilson lines is that they satisfy the property

$$\begin{aligned} W(x, y)\phi(y) - \phi(x) &\rightarrow e^{i\alpha(x)}W(x, y)e^{-i\alpha(y)}e^{i\alpha(y)}\phi(y) - e^{i\alpha(x)}\phi(x) \\ &= e^{i\alpha(x)}(W(x, y)\phi(y) - \phi(x)) \end{aligned}$$

- Taking $y^\mu = x^\mu + \delta x^\mu$, we can obtain the derivative via

$$D_\mu \phi(x) \equiv \lim_{\delta x^\mu \rightarrow 0} \frac{W(x, x + \delta x)\phi(x + \delta x) - \phi(x)}{\delta x}$$

- This gives a transformation law that we expect

$$D_\mu \phi(x) \rightarrow e^{i\alpha(x)}D_\mu \phi(x)$$

- Writing the wilson line in terms of a perturbation,

$$W(x, x + \delta x) = 1 - e\delta x^\mu A_\mu(x) + \mathcal{O}(\delta x^2)$$

- This gives, as we expect, the abelian gauge transformation

$$A_\mu(x) \rightarrow A_\mu(x) + \frac{1}{e}\partial_\mu \alpha(x)$$

- The Gauge field is a connection

- This provides the abelian covariant derivative

$$D_\mu \phi(x) = \partial_\mu \phi(x) - ieA_\mu \phi$$

- The *closed-form* for the wilson line is written as

$$W_P(x, y) = \exp\left(ie \int_y^x A_\mu(z) dz^\mu\right)$$

- Notice that this is a line integral
- This is the true Wilson Line

- Parametrizing the wilson line with $z^\mu(\lambda)$ such that $0 \leq \lambda \leq 1$ and $z^\mu(0) = y^\mu, z^\mu(1) = x^\mu$, we obtain that

$$W_P(x, y) = \exp \left(ie \int_0^1 \frac{dz^\mu(\lambda)}{d\lambda} A_\mu(z(\lambda)) d\lambda \right)$$

- We can test how this behaves under a gauge transformation

$$\begin{aligned} \blacktriangleright \quad W_P(x, y) &\rightarrow \exp \left[ie \int_y^x A_\mu(z) dz^\mu + i \int_y^x \partial_\mu \alpha(z) dz^\mu \right] \\ &= e^{i\alpha(x)} W_P(x, y) e^{-i\alpha(y)} \end{aligned}$$

- This behaves correctly from our definition of a Wilson line

- Wilson Loops:

- Setting $x = y$, we obtain that

$$W_P^{\text{loop}} = \exp \left(ie \oint_P A_\mu dx^\mu \right)$$

- Via stokes' theorem,

$$W_p^{\text{loop}} = \exp \left(i \frac{e}{2} \int_\Sigma F_{\mu\nu} d\sigma^{\mu\nu} \right) = 1 + i \frac{e}{2} \int_\Sigma F_{\mu\nu} d\sigma^{\mu\nu} + \mathcal{O}(e^2)$$

where $\sigma^{\mu\nu}$ is a surface element

- Notice that now the wilson line only depends on the field tensor $F^{\mu\nu}$ (this should not be surprising since it depends on the gauge field)
- Checking the transformation for the commutator $[D_\mu, D_\nu]$, we obtain

$$[D_\mu, D_\nu] \phi(x) \rightarrow e^{i\alpha(x)} [D_\mu, D_\nu] \phi(x)$$

- therefore, we obtain

$$[D_\mu, D_\nu] \phi(x) = ([\partial_\mu, \partial_\nu] - ie[\partial_\mu, A_\nu] + ie[\partial_\nu, A_\mu]) \phi(x) = -ieF_{\mu\nu}(x)$$

- Hence, we can define the EM field tensor as

$$F_{\mu\nu}(x) = \frac{i}{e} [D_\mu, D_\nu]$$

9.2.2 Non Abelian Wilson Lines

- We would like to repeat the same process as we did with abelian wilson lines
 - Lagrangians have more symmetry than just phase rotation in non abelian theories (by definition)
- Consider the kinetic lagrangian with N particles

$$\mathcal{L}_{kin} = \sum_{j=1}^n \bar{\psi}_j (i\not{\partial} - m) \psi_j$$

- This has a global $SU(N)$ symmetry
- Such particles transform as

$$\psi_i \rightarrow (e^{i\alpha^a T^a})_{ij} \psi_j$$

- α^a is not a function of x for global symmetries

Definition 9.2.2.1 (Non Abelian Wilson Line).

- We are now motivated to write

$$W_P(x, y) = P \left\{ \exp \left(ig \int_y^x A_\mu^a(z) T^a dz^\mu \right) \right\}$$

where P is path ordering

- The Taylor expansion of the wilson line is

$$W_P(z, y) = 1 + ig \int_0^1 \frac{dz^\mu(\lambda)}{d\lambda} A_\mu^a(z(\lambda)) T^a d\lambda \\ - \frac{1}{2} g^2 \int_0^1 d\lambda \int_0^1 d\tau \frac{dz^\mu(\lambda)}{d\lambda} \frac{dz^\nu(\tau)}{d\tau} A_\mu^a(z(\lambda)) A_\nu^b(z(\tau)) (T^a T^b \theta(\lambda - \tau) + T^b T^a \theta(\tau - \lambda)) + \dots$$

- Under a gauge transformation

$$W_P(x, y) = e^{i\alpha^a(x) T^a} W_P(x, y) e^{-i\alpha^a(y) T^a}$$

which definitely looks correct

- We are motivated to define a variable $\underline{A}_\mu \equiv A_\mu^a T^a$
 - This gives

$$W_P(x, y) = P \left\{ \exp \left(ig \int_{y-x}^x \underline{A}_\mu(z) dz^\mu \right) \right\}$$

- An infinitesimal transformation for W is

$$W(x^\mu, x^\mu + \delta x^\mu) = 1 + i \epsilon \underline{A}_\mu \delta x^\mu$$

- Local transformations can be expressed in terms of

$$U(x) = e^{i\alpha^a(x) T^a}$$

- In $SU(N)$, $U^\dagger(y) = U(y)$

- Then,

$$\vec{\psi} = U(x) \cdot \vec{\psi}(x)$$

$$W(x, y) \rightarrow U(x)W(x, y)U^\dagger(y)$$

- We would like to analyze what happens to the gauge field under such a transformation
 - Consider how the covariant derivative transforms:

$$D_\mu \vec{A} \rightarrow U \cdot D_\mu \vec{A}$$

- Therefore, we have that

$$(\partial_\mu - ig \underline{A'_\mu})U\psi = U(\partial_\mu - ig \underline{A_\mu})\psi$$

where $\underline{A_\mu}$ is the transformed A_μ

- Therefore, the transformation for A_μ^a is

$$A_\mu^a(x) \rightarrow A_\mu^a(x) + \frac{1}{g} \partial_\mu \alpha^a(x) - f^{abc} \alpha^b(x) A_\mu^c + O(\alpha^2)$$

- The covariant derivative commutator transforms as

$$[D_\mu, D_\nu]\psi(x) = (-ig(\partial_\mu \underline{A_\nu} - \partial_\nu \underline{A_\mu}) - g^2[\underline{A_\mu}, \underline{A_\nu}])\psi(x)$$

- This allows for a definition of the *gluon field tensor*:

$$\underline{F_{\mu\nu}} \equiv \frac{i}{g} [D_\mu, D_\nu] = (\partial_\mu \underline{A_\nu} - \partial_\nu \underline{A_\mu}) - ig[\underline{A_\mu}, \underline{A_\nu}]$$

$$\underline{F_{\mu\nu}} = F_{\mu\nu}^a T^a$$

-

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c$$

- The transformation law for the field tensor is

$$F_{\mu\nu}^a \rightarrow F_{\mu\nu}^a - f^{abc} \alpha^b F_{\mu\nu}^c$$

which is the same whether or not α is global or local

- This is because

$$F_{\mu\nu}^\alpha = -F_{\nu\mu}^\alpha$$

- Kinetic terms in the lagrangian only depend on $F_{\mu\nu}^a$

- The $SU(N)$ lagrangian is

$$\mathcal{L} = -\frac{1}{4}(F_{\mu\nu}^a)^2 + \sum_{i,j=1}^N \bar{\psi}_i (i\delta_{ij}\not{\partial} + gA_\mu^a T_{ij}^a - m\delta_{ij})\psi_j$$

- In this lagrangian, g acts as $-e$
- There is an extra term that is consistent with gauge invariance that can be added to this lagrangian:

$$\mathcal{L}_\theta = \theta \varepsilon^{\mu\nu\alpha\beta} F_{\mu\nu}^\alpha F_{\alpha\beta}^\alpha = 2\theta \partial_\mu (\varepsilon^{\mu\nu\alpha\beta} A_\nu^\alpha F_{\alpha\beta}^\alpha)$$

- Notice that this is a total derivative, hence it does not contribute perturbatively
- There should be non perturbative effects that this term adds (the lack of them is known as the *strong CP problem*)
- I claim that there is no conserved current for this lagrangian
- Consider the expansion of the lagrangian

$$\mathcal{L} = -\frac{1}{4}(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c)^2 + \bar{\psi}_i (i\delta_{ij}\gamma^\mu \partial_\mu + g\gamma^\mu A_\mu^a T_{ij}^a - m\delta_{ij})\psi_j$$

- The EoM for this lagrangian are

$$\partial_\mu F_{\mu\nu}^a + gf^{abc}A_\mu^b F_{\mu\nu}^c = -g\bar{\psi}_i \gamma_\nu T_{ij}^a \psi_j \quad (\text{Gauge})$$

and

$$-(i\not{\partial} - m)\psi_i = -gA_\mu^a T_{ij}^a \psi_j \quad (\text{Ferm.})$$

- The global symmetries for this are

$$\begin{aligned} \psi_i &\rightarrow \psi_i + i\alpha^a T_{ij}^a \psi_j \\ A_\mu^a &\rightarrow A_\mu^a - f^{abc}\alpha^b A_\mu^c \end{aligned}$$

- By Noether's theorem,

$$J_\mu = \sum_n \frac{\partial L}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha}$$

- In a non abelian theory, there are $N^2 - 1$ currents
 - Summing over ψ_i and A_μ^a provides

$$J_m^\mu = -\bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j + f^{abc} A_\nu^b F_c^{\mu\nu}$$

- Notice, however, that this current is gauge invariant.

- Hence, it is not physical
- We can try and create a matter current

$$j_\mu^a = -\bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j$$

- This satisfies

$$D_\mu j_\mu^a = 0$$

- It Still doesnt satisfy $\partial_\mu j_\mu^a = 0$

Theorem 9.2.2.2 (WEINBERG WITTEN THEOREM).

- *A non abelian symmetry for massless spin 1 implies that there is no gauge invariant conserved charge*

9.2.3 The Gluon Propagator

9.2.3.1 F-P ghosts

- To Start, let us consider the equations of motion for a photon with an external current

$$(g_{\mu\nu} \square - \partial_\mu \partial_\nu) A_\nu = J_\mu$$

- This is not invertible, similar to the gluon equation of motion

$$f(\xi) = \int \mathcal{D}\pi e^{-i \int d^4x \frac{1}{2\xi} (D\pi)^2} = \int \mathcal{D}\pi e^{-i \int d^4x \frac{1}{2\xi} (D\pi - \partial_\mu A_\mu)^2}$$

- This is independed of A_μ since we can take $\pi \rightarrow \pi - \frac{1}{\square} \partial_\mu A_\mu$
- Therefore, we obtained

$$\begin{aligned} \int \mathcal{D}A_\mu \mathcal{D}\phi_i e^{i \int d^4x L[A, \phi_i]} &= \frac{1}{f(\xi)} \int \mathcal{D}\pi \mathcal{D}A_\mu \mathcal{D}\phi_i e^{i \int d^4x (L(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A_\mu)^2)} \\ &= \left[\frac{1}{f(\xi)} \int \mathcal{D}\pi \right] \int \mathcal{D}A_\mu \mathcal{D}\phi_i e^{i \int d^4x (L[A, \phi_i] - \frac{1}{2\xi} (\partial_\mu A_\mu)^2)} \end{aligned}$$

in QED

- We want to do a similar thing for QCD
- The gauge transformation is

$$A_\mu^a \rightarrow A_\mu^a + \frac{1}{g} \partial_\mu \alpha^a + f^{abc} A_\mu^b \pi^c$$

- Notice that there are $N^2 - 1$ π^a fields due to them transforming in the adjoint representation
- Since π^a transforms in the adjoint rep we can write

$$A_\mu^a \rightarrow A_\mu^a + \frac{1}{g} D_\mu \pi^a$$

where the adjoint covar. derivative is

$$D_\mu \pi^a = \partial_\mu \pi^a + g f^{abc} A_\mu^b \pi^c$$

- We now have the machinery to write a functional

$$f[A] = \int D\pi \exp \left[-i \int d^4x \frac{1}{2\xi} (\partial_\mu D_\mu \pi^a)^2 \right]$$

“Tickle Tip”

- Define $\alpha^a[A]$ s.t.

$$\partial_\mu A_\mu^a = \frac{1}{g} \partial_\mu D_\mu^a \alpha^a[A]$$

- Therefore,

$$f[A] = \int D\pi \exp \left[-i \int d^4x \frac{1}{2\xi} (\partial_\mu A_\mu^\alpha - \partial_\nu A_\mu^\alpha)^2 \right]$$

- This allows us to obtain the full path integral

$$\begin{aligned} & \int DAD\phi_i e^{i \int d^4x L(A, \phi_i)} \\ &= \int D\pi DAD\phi_i \frac{1}{f[A]} \exp \left(i \int d^4x L(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A^\mu - \partial_\mu A^\mu)^2 \right) \\ &= \left(\int D\pi \right) \int DAD\phi_i \frac{1}{f[A]} \exp \left(i \int d^4x L(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A^\mu)^2 \right) \end{aligned}$$

- We now wish to rewrite $f[A]$ (?)

$$\triangleright f = \sqrt{\frac{1}{\det(\partial_\mu D_\mu)^2}} \times \text{const.}$$

- Then,

$$\begin{aligned} Z[0] &= \text{const} \times \int DA_m D\phi_i (\det(\mathcal{M}_{mn})) \\ &\times \exp \left\{ i \int d^4x \left[\mathcal{L}(A, \phi_i) - \frac{1}{2\xi} (\partial_\mu A_m^a)^2 \right] \right\} \end{aligned}$$

- We can use identities for gaussians to write

$$\det(\partial^\mu D_\mu) = \int \mathcal{D}\bar{c}\mathcal{D}c \exp\left(i \int d^4x \bar{c}(-\partial^\mu D_\mu)c\right)$$

- The identity involved in this is

$$\det(\mathcal{O}) = \int \mathcal{D}\bar{\psi}\mathcal{D}\psi \exp\left(-i \int \bar{\psi}\mathcal{O}\psi\right)$$

- Hence, we obtain the result that

$$Z[c] = \text{const.} \times \int \mathcal{D}A_\mu \mathcal{D}\phi \mathcal{D}\bar{c}\mathcal{D}c \\ \times \exp\left\{i \int d^4x \left[L(A, \phi) - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 - \bar{c}^a \partial^\mu D_\mu c^a\right]\right\}$$

- We have obtained a very bad result:
 - In QCD, as a result of trying to find a gluon propagator, we find these annoying things called Fadeev-Popov ghosts and antighosts
 - They pop up out of nowhere and are not gauge invariant
 - They are aphysical clearly
- The Lagrangian must (and I mean must unless you are willing to work in a obnoxious gauge) be written with these in mind

$$\mathcal{L}_{R_\xi} = -\frac{1}{4}(F_{\mu\nu}^a)^2 - \frac{1}{2\xi}(\partial_\mu A^{\mu a})^2 + (\partial_\mu \bar{c}^a)(\delta^{ac}\partial^\mu + g f^{abc} A^{\mu b})c^c$$

- This is called the Fadeev-Popov lagrangian
- Luckily, we also have the gluon propagator

$$\text{gluon line} = i \frac{-g^{\mu\nu} + (1-\xi)\frac{p^\mu p^\nu}{p^2}}{p^2 + i\varepsilon} \delta^{ab}$$

- Observe that

$$1 = \int \mathcal{D}\pi \delta(G(A_\mu^a - D_\mu \pi^a)) \det\left(\frac{\delta G[A_\mu^a - D_\mu \pi^a]}{\delta \pi^a}\right)$$

- $G[A]$ is a functional which represents our choice of gauge
- Then,

$$\det\left[\frac{\delta G\{A_m^a - D_m \sigma^a\}}{\delta \sigma^b}\right] = \det(\partial^m D_m) = \frac{1}{f[A]} = \int \mathcal{D}\bar{c}\mathcal{D}c \exp\left(i \int d^4x \bar{c}^a(-\partial^m D_m)c^a\right)$$

- Multiplying $Z[0]$ by 1 yields

$$Z[0] = \text{const} \times \int \mathcal{D}\pi \int \mathcal{D}A_m \mathcal{D}\phi_i \delta(G[A_m^a - D_m \pi^a]) \\ \times \det\left(\frac{\delta G[A_m^a - D_m \pi^a]}{\delta \pi^b}\right) \exp\left(i \int d^4x \mathcal{L}[A, \phi]\right)$$

- We use the trick of shifting

$$A_\mu^a \rightarrow A_\mu^a + D_\mu \pi^a$$

due to gauge invariance

- Our result is that

$$\Rightarrow Z[0] = \text{const.} \left(\int D\pi \right) \int DA_\mu D\phi_i \delta(G[A_\mu^a]) \\ \times \det\left(\frac{\delta G[A_\mu^a - D_\mu \pi^a]}{\delta \pi^b}\right) \times \exp\left(i \int d^4x L[A, \phi_i]\right)_{\pi \rightarrow 0}$$

- Shifting G by a constant keeps the determinant the same. Therefore, we can average over a gaussian which yields

$$\int \mathcal{D}\chi \exp\left(-i \int d^4x \frac{X^2}{2\xi}\right) \delta(G[A_m^a] - \chi) = \exp\left(-i \int d^4x \frac{1}{2\xi} G[A_m^a]^2\right)$$

- This yields our final form of $Z[0]$

$$Z[0] = \text{const} \times \int \mathcal{D}A_\mu \mathcal{D}\phi_i \det\left(\frac{\delta G[A_\mu, \phi_i]}{\delta \phi_j}\right) \times \exp\left[i \int d^4x \left(L[A, \phi_i] - \frac{1}{2g} G[A_\mu]^2\right)\right]$$

9.2.3.2 BRST Invariance

- There is another symmetry that the lagrangian is invariant under.
 - This symmetry involves grassmann numbers and qcd ghosts
- Consider the lagrangian

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 + (D_\mu \phi_i^*)(D^\mu \phi_i) - m^2 \phi_i^* \phi_i - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 - \bar{c}\square c$$

- If we define a shift

$$\phi_i(x) \rightarrow e^{i\theta(x)} \phi_i(x) \approx \phi_i(x) + i\theta(x) \phi_i(x) \\ A_\mu(x) \rightarrow A_\mu(x) + \frac{1}{e} \partial_\mu \theta(x)$$

and either use the EOM or take

$$(\partial_\mu A_\mu)^2 \rightarrow (\partial_\mu A_\mu)^2 + \frac{2}{e}(\partial_\mu A_\mu)(\theta \square c) + \frac{1}{e^2}(\theta \square c)(\theta \square c)$$

$$\bar{c}(x) \rightarrow \bar{c}(x) - \frac{1}{e}\theta \frac{1}{\xi} \partial_\mu A^\mu(x)$$

, this lagrangian is invariant

- This gives us **BRST Invariance**

Definition 9.2.3.2.1 (BRST Invariance).

- BRST Invariance is a generalization of gauge invariance for the Fadeev Popov lagrangian
- This works in the non-abelian case similarly, where the lagrangian is

$$\mathcal{L}_{FP} = \mathcal{L}[A_\mu^a, \phi_i] - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + (\partial_\mu \bar{c}^a)(D_\mu c^a)$$

- One can define transformations

$$\phi_i \rightarrow \phi_i + i\theta^a T_{ij}^a \phi_j$$

$$A_\mu^a \rightarrow A_\mu^a + \frac{1}{g}\theta D_\mu c^a$$

$$\bar{c}^a \rightarrow \bar{c}^a - \frac{1}{g}\theta \frac{1}{\xi} \partial_\mu A_\mu^a$$

which leave everything but D_μ invariant. Hence. we need a final transformation which, after a lot of algebra, results in

$$D_\mu c^a \rightarrow D_\mu c^a + \theta f^{abc}(D_\mu c^b)c^c$$

$$c^a \rightarrow c^a - \frac{1}{2}\theta f^{abc}c^b c^c$$

- or alternatively, after more algebra,

$$D_\mu c^a \rightarrow D_\mu c^a + g f^{abc}(D_\mu c^b)c^c - g f^{abc} \left[\frac{1}{2}(\partial_\mu c^b)c^c + \frac{1}{2}c^b(\partial_\mu c^c) + \frac{g}{2}A_\mu^b f^{cde}c^d c^e \right]$$

9.2.3.3 Axial gauges

Summary

- Axial gauges are not very useful to work with but they remove the ghosts that are inconvenient to work with
- The gluon propagator in an axial gauge is

$$i\Pi_{\text{lightcone}}^{\mu\nu ab} = \frac{i}{p^2 + i\epsilon} \left[-ig^{\mu\nu} + \frac{r^\mu p^\nu + p^\mu r^\nu}{rp} \right] \delta^{ab}$$

which, as you may be able to see, is not the most fun to work with

- The most general lagrangian that we can write to gauge fix our other one is

$$-\frac{1}{2\lambda}(\partial^\mu A_\mu^a)^2 + \bar{c}^a \partial^\mu (\delta^{ab} \partial_\mu + g f^{abc} A_\mu^b) c^c$$

with two parameters: λ , a scalar, and r^μ , a 4-vector.

- The general propagator in this axial gauge is

$$\Pi_{axial}^{\mu\nu ab} = \frac{i}{p^2 + i\epsilon} \left[-g^{\mu\nu} + \frac{r^\mu p^\nu + r^\nu p^\mu}{r \cdot p} - \frac{(r^2 + \lambda p^2) p^\mu p^\nu}{p^2} \right] \delta^{ab}$$

- We of course want this to satisfy the ward identity, but we do not have to change anything since it already does.
- In the *Lightcone gauge*, we force that $r^2 = 0$, $\lambda = 0$, which yields our propagator

9.3 Lattice QCD

Summary

- Lattice QCD works by discretizing standard QCD into plaquettes and evaluating wilson loops on them
- Fields are at specific lattice sites

Definition 9.3.1 (Discrete Wilson line).

- First, lets consider some notation. $\hat{\mu}$ and $\hat{\nu}$ will be unit vectors (of length a since in lattice qcd, everything is scaled by such a factor) in the μ and ν directions. $W_\mu(n)$ will denote a wilson line at some site n and some other site $n + \hat{\mu}$
- These wilson lines will transform under the law

$$W_\mu(n) \rightarrow U(n) W_\mu(n) U^\dagger(n + \hat{\mu})$$

because $\phi(n) \rightarrow U(n) \phi(n)$

Verification:

- Just testing that our wilson lines behave correctly (as they should),

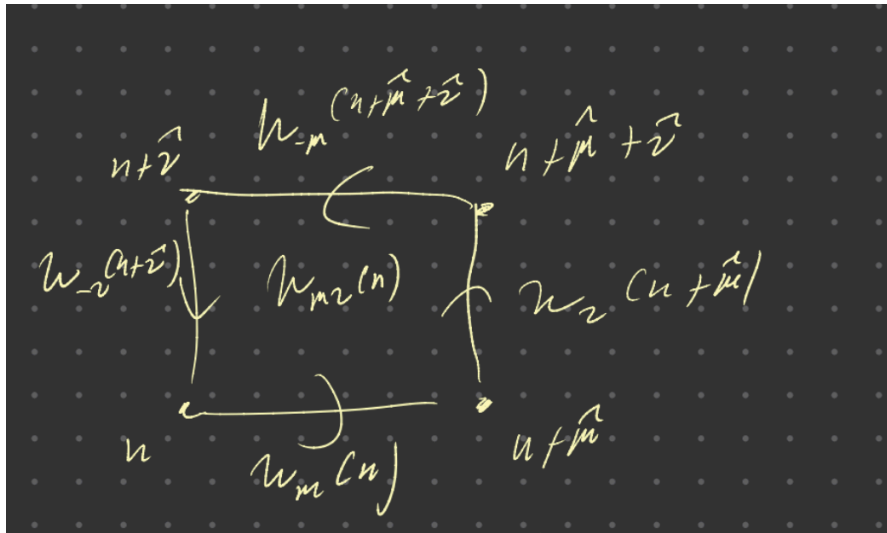
$$\begin{aligned} \vec{\phi}^\dagger(n) W_\mu(n) \vec{\phi}(n + \hat{\mu}) &\rightarrow \vec{\phi}^\dagger(n) U^\dagger(n) U(n) W_\mu(n) \\ &\times U^\dagger(n + \hat{\mu}) U(n + \hat{\mu}) \vec{\phi}(n + \hat{\mu}) = \vec{\phi}^\dagger(n) W_\mu(n) \vec{\phi}(n + \hat{\mu}) \end{aligned}$$

Definition 9.3.2 (Backwards Wilson Line).

- $W_{-mu}(n) = W_{\mu}^{\dagger}(n - \hat{\mu})$

Definition 9.3.3 (plaquette).

- something of the form in this picture



- It is evaluated as

$$W_{\mu\nu}(N) \equiv W_{-\nu}(n + \hat{\nu})W_{-\mu}(n + \hat{\mu} + \hat{\nu})W_{\nu}(n + \hat{\mu})W_{\mu}(N)$$

- If we write

$$W_{\mu}(n) = \exp(ia\mathbf{A}_{\mu}(n))$$

, we can see how this would be related to the continuous version

Evaluating Plaquettes:

- How do we evaluate plaquettes?

- ▶ Consider the *Campbell Baker Hausdorff* formula:

$$\exp(A)\exp(B) = \exp\left(A + B + \frac{1}{2}[A, B] + \dots\right)$$

- Applying this to our definition of $W_{\mu\nu}$,

$$\begin{aligned}
\ln W_{m\nu}(N) &= ia(\underline{A}_\mu(n) + \underline{A}_\nu(n + \hat{\mu}) - \underline{A}_\mu(n + \hat{\nu}) - \underline{A}_\nu(n)) \\
&+ \frac{a^2}{2} \{ [\underline{A}_\nu(N) + \underline{A}_\mu(n + \hat{\nu}), \underline{A}_\nu(n + \hat{\mu}) - \underline{A}_\mu(n)] \\
&- [\underline{A}_\nu(N), \underline{A}_\mu(n + \hat{\nu})] - [\underline{A}_\nu(n + \hat{\mu}), \underline{A}_\mu(n)] \} \\
&+ \mathcal{O}(a^3)
\end{aligned}$$

- Via a Taylor expansion,

$$\underline{A}_\nu(n + \hat{\mu}) = \underline{A}_\nu(n) + a\partial_\mu \underline{A}_\nu(n) + \mathcal{O}(a^2)$$

,

$$\begin{aligned}
W_{\mu\nu}(n) &= \exp\{ia^2(\partial_\mu \underline{A}_\nu(n) - \partial_\nu \underline{A}_\mu(n)) + a^2[\underline{A}_\mu(n), \underline{A}_\nu(n)] + \mathcal{O}(a^3)\} \\
&= \exp\{ia^2 \underline{F}_{\mu\nu}(n) + \mathcal{O}(a^3)\}
\end{aligned}$$

- Consider the Yang Mills action:

$$S_{YM}[F_{\mu\nu}] = i \int d^4x \left(-\frac{1}{4g^2} (F_{\mu\nu}^a)^2 \right)$$

- We would like to discretize this

- At small a ,

$$W_{\mu\nu}(N) = \mathbb{I} + ia^2 \underline{F}_{\mu\nu}(N) - \frac{a^4}{2} \underline{F}_{\mu\nu}^2(N) + \mathcal{O}(a^6)$$

- Additionally, we can discretize the YM action by writing

$$S_{YM}[F_{\mu\nu}] = i \int d^4x \left(-\frac{1}{4g^2} (F_{\mu\nu}^a)^2 \right) = \frac{-ia^4}{4Ng^2} \sum_{n,\mu\nu} \text{tr}(F_{\mu\nu}^2)$$

- We finally obtain the lattice action:

$$S_{\text{lattice}}[W_{\mu\nu}] = \frac{-i}{2g^2 N} \sum_{\nu,\mu} \text{Re}(\text{tr}(\mathbb{I} - W_{\mu\nu}(N)))$$

- For corr. functions, you would use this action in a path integral like

$$C(x) \langle \Omega | \mathcal{O}(0) \mathcal{O}(x) | \Omega \rangle = \int \mathcal{D}A_\mu \mathcal{D}\bar{u} \mathcal{D}u e^{iS} \mathcal{O}(0) \mathcal{O}(x)$$

However, it would give a rapid phase shift. Instead, you must perform a *wick rotation*

9.4 QCD Feynman Rules

- Now, we have the full lagrangian for QCD (the F-P) one allowing us to read off feynman rules

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4}(F_{\mu\nu}^a)^2 - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + \\ & (\partial_\mu \bar{c}^a)(\delta^{ac}\partial_\mu + g f^{abc} A_\mu^b)c^c + \bar{\psi}_i(i\not{\partial}\delta_{ij} + g A_\mu^a T_{ij}^a - m\delta_{ij})\psi_j + \\ & [(\delta_{ki}\partial_\mu - ig A_\mu^a T_{ki}^a)\phi_i]^\dagger [(\delta_{kj}\partial_\mu - ig A_\mu^a T_{kj}^a)\phi_j] - m^2\phi_i^\dagger\phi_i \end{aligned}$$

- The kinetic terms for the lagrangian are

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4}(\partial_\mu A_n^a - \partial_\nu A_\mu^a)^2 - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + \bar{\psi}_i(i\not{D} - m)\psi_i - \phi_i^*(\square + m^2)\phi_i - \bar{c}^a \square c^a$$

- The rules are as follows:

$$\mu; a \text{---}\text{ooooo} \nu; b = \frac{-g^{\mu\nu} + (1 - \xi) \frac{p^\mu p^\nu}{p^2}}{p^2} \delta^{ab}$$

- For ghosts,

$$\text{=====}\blacktriangleright\text{=====} = \frac{\delta^{ab}}{p^2 + i\epsilon}$$

- For colored fermions,

$$i \longrightarrow j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}$$

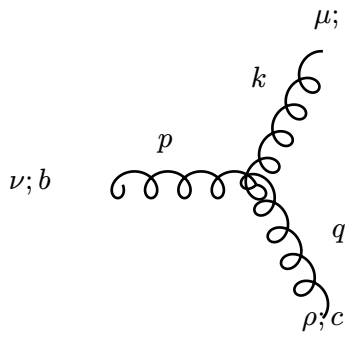
- For colored scalars,

$$i \text{ --- } \blacktriangleright \text{ --- } j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}$$

- The interaction terms (which are where this gets interesting) are the following:

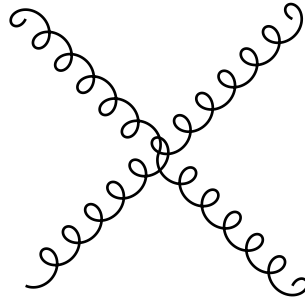
$$\begin{aligned}\mathcal{L}_{int} = & -gf^{abc}(\partial_\mu A_\nu^a)A_\mu^b A_\nu^c - \frac{1}{4}g^2(f^{ab}A_\mu^a A_\nu^b)(f^{cd}A_\mu^c A_\nu^d) \\ & + gf^{abc}(\partial_\mu \bar{c}^a)A_\mu^b c^c + gA_\mu^a \bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j \\ & + igA_\mu^a T_{ij}^a (\phi_i \overleftrightarrow{\partial}^\mu \phi_j) + g^2 \phi_i^* A_\mu^a T_{ik}^a T_{kj}^b A_\mu^b \phi_j\end{aligned}$$

- Trips g:

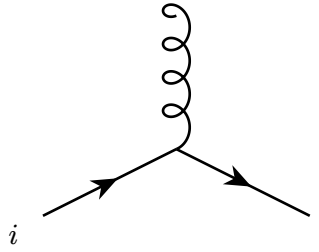


$$= g A^{abc} [g^{\mu\nu} (k - p)^\mu + g^{\nu\rho} (p - q)^\mu + g^{\rho\mu} (q - k)^\nu]$$

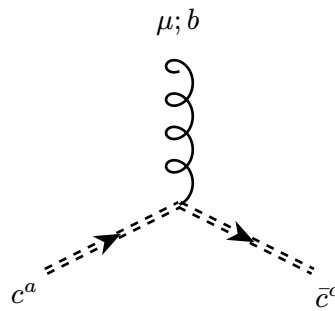
- Quads g:



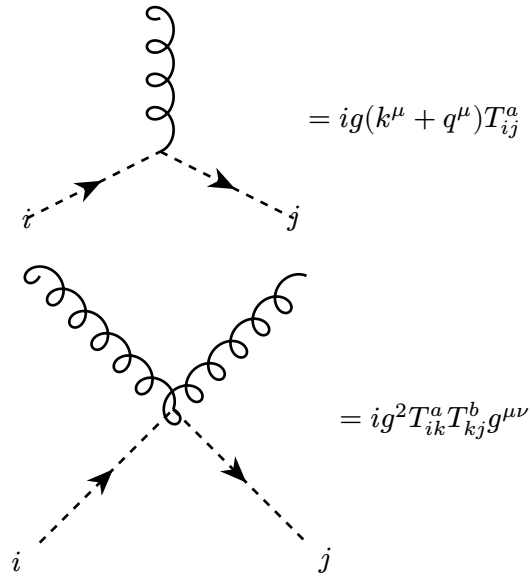
$$= -ig^2 [f^{abe} f^{cde} (g^{\mu\rho} g^{\nu\sigma} - g^{\mu\sigma} g^{\nu\rho}) + f^{ace} f^{bde} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\sigma} g^{\nu\rho}) + f^{ade} f^{bce} (g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma})]$$



$$= ig\gamma^\mu T_{ij}^a$$



$$= -gf^{abc} p^\mu$$



9.5 Helicity Spinors

Summary:

- Helicity spinors are a way to write polarizations ε_μ and momentum p_μ in terms of a more natural way (bispinors)
 - This is more convenient specifically for calculating gluon scattering
- They are written in the weyl representation where ψ_α and $\tilde{\chi}^{\dot{\alpha}}$ are left and right handed spinors
- Notice that spin 1 fields transform in the $(\frac{1}{2}, \frac{1}{2})$ representation of the Lorentz group
 - This entails that we can write them as bispinors instead of 4-vectors
- Remember the mathematical relations for spinors in the weyl representation:
 - $\varepsilon^{\alpha\beta}\psi_\alpha(x)\psi_\beta(x)$ and $\varepsilon_{\dot{\alpha}\dot{\beta}}\tilde{\psi}^{\dot{\alpha}}\tilde{\psi}^{\dot{\beta}}$ are L.I.
- Some useful relations with these (and the pauli matrices) are

$$g^{\mu\nu}\sigma_\mu^{\alpha\dot{\alpha}}\sigma_\nu^{\beta\dot{\beta}} = 2\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}$$

$$\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}\sigma^{\mu\dot{\beta}\beta} = \bar{\sigma}^\mu_{\dot{\alpha}\alpha}$$

- For the inner product, defined as $\psi\chi \equiv \varepsilon^{\alpha\beta}\psi_\alpha(x)\chi_\beta(x)$
 - $\psi\chi = \chi\psi$
- All of these spinors are fields. What happens when we use constant spinors?

Definition 9.5.1 (Helicity Spinor).

- $\mathbb{R}$ or $\mathbb{C}$ doublets that transform in the $(\frac{1}{2}, 0)$ or $(0, \frac{1}{2})$ representation of the Lorentz Group
- The inner products for these are
- $\langle \lambda \chi \rangle = \epsilon^{\alpha\beta} \lambda_\alpha \chi_\beta = \lambda_\alpha \chi^\alpha = -\lambda^\alpha \chi_\alpha$
- $[\lambda \chi] = \epsilon_{\alpha\beta} \tilde{\lambda}^\alpha \tilde{\chi}^\beta = \tilde{\lambda}^\alpha \tilde{\chi}_\alpha = -\tilde{\chi}_\alpha \tilde{\lambda}^\alpha$
- These products have some identities due to them commuting:

$$\begin{aligned}\langle \lambda \chi \rangle &= -\langle \chi \lambda \rangle, [\lambda \chi] = -[\chi \lambda] \\ [\lambda \lambda] &= \langle \lambda \lambda \rangle = 0\end{aligned}$$

9.5.1 Momentum

- For bispinor representation of momentum, we have:

$$p_{\alpha\dot{\alpha}} \equiv \sigma_{\alpha\dot{\alpha}}^\mu p_\mu = \begin{pmatrix} p^0 - p^3 & -p^1 + ip^2 \\ -p^1 - ip^2 & p^0 + p^3 \end{pmatrix}$$

- **Relations** with these spinors are:

$$\begin{aligned}p^{\alpha\dot{\alpha}} &\equiv \sigma_{\alpha\dot{\alpha}}^\mu p_\mu, \quad p_{\alpha\dot{\alpha}} = \bar{\sigma}_{\alpha\dot{\alpha}}^\mu p_\mu, \\ p^\mu &= \frac{1}{2} \bar{\sigma}^{\mu\alpha\dot{\alpha}} p_{\alpha\dot{\alpha}}, \quad p^\mu = \frac{1}{2} \bar{\sigma}_{\alpha\dot{\alpha}}^\mu p^{\alpha\dot{\alpha}}.\end{aligned}$$

- If $p^{\alpha\dot{\alpha}}$ is lightlike (that is, $\det(p^{\alpha\dot{\alpha}}) = 0$), we can express it in terms of an outer product

$$\begin{aligned}p^{\alpha\dot{\alpha}} &= \lambda^\alpha \tilde{\chi}^{\dot{\alpha}} \\ \lambda_\alpha &= \frac{z}{\sqrt{p^0 - p^3}} \begin{pmatrix} p^0 - p^3 \\ -p^1 - ip^2 \end{pmatrix}, \\ \tilde{\lambda}^{\dot{\alpha}} &= \frac{z^1}{\sqrt{p^0 - p^3}} (p^0 - p^3, -p^1 + ip^2)\end{aligned}$$

9.5.1.1 Little Group

- These momenta bispinors are invariant under $\lambda_\alpha \rightarrow z \lambda_\alpha$ and $\tilde{\lambda}^{\dot{\alpha}} \rightarrow \frac{1}{z} \tilde{\lambda}^{\dot{\alpha}}$
 - This is equivalent to a phase shift $z = e^{i\phi}$ if the momenta are real

9.5.1.2 Connecting Momentum and Helicity Spinors

- Notice that the dot product

$$\begin{aligned}
p \cdot q &= \frac{1}{4} g_{\mu\nu} \sigma_{\alpha\dot{\alpha}}^{\mu} \sigma_{\beta\dot{\beta}}^{\nu} \lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}} \chi^{\beta} \tilde{\chi}^{\dot{\beta}} \\
&= \frac{1}{2} \varepsilon_{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} \lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}} \chi^{\beta} \tilde{\chi}^{\dot{\beta}} = \frac{1}{2} \langle \chi \lambda \rangle [\chi \lambda]
\end{aligned}$$

- Therefore, we obtain that $\langle \chi \lambda \rangle$ and $[\chi \lambda]$ are the square roots of inner products up to a phase shift

$$\begin{aligned}
\langle \lambda \chi \rangle &= \sqrt{2p \cdot q} e^{i\phi} \\
[\chi \lambda] &= \sqrt{2p \cdot q} e^{-i\phi}
\end{aligned}$$

- This allows us to write

$$\lambda_{\alpha} = p\rangle, \quad \lambda_{\alpha} = \langle p, \quad \tilde{\lambda}^{\dot{\alpha}} = [p, \quad \tilde{\lambda}_{\dot{\alpha}} = p]$$

As for our momentum,

$$p^{\alpha\alpha} = p\rangle[p, p_{\alpha\alpha} = p]\langle p$$

- The dot products of momenta are

$$\begin{aligned}
q \cdot p &= q^{\mu} p_{\mu} = \frac{1}{2} q_{\alpha\dot{\alpha}} p^{\dot{\alpha}\alpha} \\
&= \frac{1}{2} \text{tr}\{q\} \langle qp \rangle [p] \\
&= \frac{1}{2} \langle qp \rangle [qp]
\end{aligned}$$

- Here is another identity that is useful:

$$\begin{aligned}
\sum_i p_i &= 0 \\
\sum_j \lambda_{\alpha}^j \lambda_{\alpha}^j &= 0 \\
\sum_{j=1}^n j \rangle [j=1] [1+2] [2+\dots+n] [n=0]
\end{aligned}$$

- To help us with being terse, we write $i\rangle \equiv p_i\rangle$, etc.

- Our conservation of momentum

$$\sum_j \langle ij \rangle [jk] = 0$$

- Since the spinors are two dimensions, we can write them in terms of two other spinors (a *linear combination*)

$$1\rangle = \frac{\langle 13 \rangle}{\langle 23 \rangle} 2\langle 2 - \frac{\langle 12 \rangle}{\langle 23 \rangle} 3\rangle$$

9.5.2 Polarizations

- All polarizations satisfy the relation that

$$\varepsilon_\mu^* \varepsilon^\mu = -1, \quad p_\mu \varepsilon^\mu = 0$$

- Notice that $\varepsilon_\mu \varepsilon^\mu = 0$

Example 9.5.2.1 (Helicity Polarizations).

- If the momentum is $p^\mu = (E, 0, 0, E)$ then we obtain polarizations

$$\epsilon_+^\mu = \frac{1}{\sqrt{2}}(0, 1, i, 0),$$

$$\epsilon_-^\mu = \frac{1}{\sqrt{2}}(0, 1, -i, 0)$$

- as seen before, they are the *helicity polarizations*

Definition 9.5.2.2 (Reference Momentum).

- In order to write the polarizations in terms of the momentum, we need a reference momentum. In the case of gluon scattering which this is a very useful tool for, we take the reference momentum to be the momentum of the other gluons.
- This is denoted by r^μ
- Writing

$$p^{\alpha\dot{\alpha}} = p\rangle[p, \quad r^{\alpha\dot{\alpha}} = r\rangle[r$$

- Thus, the polarizations in the bispinor format is written as

$$[\varepsilon_p^-(v)]^{\alpha\dot{\alpha}} = \sqrt{2} \frac{p\rangle[r}{[pr]}$$

$$[\varepsilon_p^+(v)]^{\alpha\dot{\alpha}} = \sqrt{2} \frac{r\rangle[p}{\langle rp\rangle}$$

$$\begin{aligned} \varepsilon_p^-(v) \cdot \varepsilon_p^+(v) &= \frac{1}{2} [\varepsilon_p^-(v)]_{\dot{\alpha}\alpha} [\varepsilon_p^+(v)]^{\alpha\dot{\alpha}} \\ &= \frac{1}{2} \frac{2}{[pr]\langle rp\rangle} \text{tr}\{r\rangle\langle pr\rangle[p] = -1 \end{aligned}$$

- since $\langle pp\rangle[pp] = 0$, we have that $\varepsilon^+ \varepsilon^+ = \varepsilon^- \cdot \varepsilon^- = 0$

9.6 QCD Feynman Rules

- Now, we have the full lagrangian for QCD (the F-P) one allowing us to read off feynman rules

$$\begin{aligned} \mathcal{L} = & -\frac{1}{4}(F_{\mu\nu}^a)^2 - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + \\ & (\partial_\mu \bar{c}^a)(\delta^{ac}\partial_\mu + g f^{abc} A_\mu^b)c^c + \bar{\psi}_i(i\not{\partial}\delta_{ij} + g A_\mu^a T_{ij}^a - m\delta_{ij})\psi_j + \\ & [(\delta_{ki}\partial_\mu - ig A_\mu^a T_{ki}^a)\phi_i]^\dagger [(\delta_{kj}\partial_\mu - ig A_\mu^a T_{kj}^a)\phi_j] - m^2\phi_i^\dagger\phi_i \end{aligned}$$

- The kinetic terms for the lagrangian are

$$\mathcal{L}_{\text{kin}} = -\frac{1}{4}(\partial_\mu A_n^a - \partial_\nu A_\mu^a)^2 - \frac{1}{2\xi}(\partial_\mu A_\mu^a)^2 + \bar{\psi}_i(i\not{D} - m)\psi_i - \phi_i^*(\square + m^2)\phi_i - \bar{c}^a \square c^a$$

- The rules are as follows:

$$\mu; a \text{---}\text{ooooo} \nu; b = \frac{-g^{\mu\nu} + (1 - \xi) \frac{p^\mu p^\nu}{p^2}}{p^2} \delta^{ab}$$

- For ghosts,

$$\text{=====}\blacktriangleright\text{=====} = \frac{\delta^{ab}}{p^2 + i\epsilon}$$

- For colored fermions,

$$i \longrightarrow j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}$$

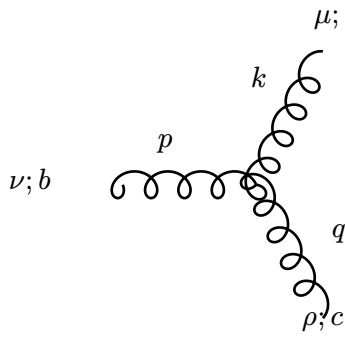
- For colored scalars,

$$i \text{ --- } \blacktriangleright \text{ --- } j = \frac{i\delta^{ij}}{\not{p} - m + i\epsilon}$$

- The interaction terms (which are where this gets interesting) are the following:

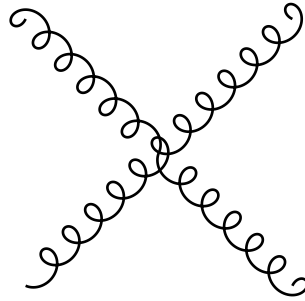
$$\begin{aligned}\mathcal{L}_{int} = & -gf^{abc}(\partial_\mu A_\nu^a)A_\mu^b A_\nu^c - \frac{1}{4}g^2(f^{ab}A_\mu^a A_\nu^b)(f^{cd}A_\mu^c A_\nu^d) \\ & + gf^{abc}(\partial_\mu \bar{c}^a)A_\mu^b c^c + gA_\mu^a \bar{\psi}_i \gamma^\mu T_{ij}^a \psi_j \\ & + igA_\mu^a T_{ij}^a (\phi_i \overleftrightarrow{\partial}_\mu \phi_j) + g^2 \phi_i^* A_\mu^a T_{ik}^a T_{kj}^b A_\mu^b \phi_j\end{aligned}$$

- Trips g:

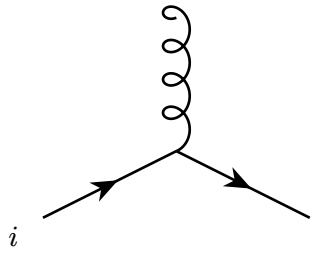


$$= gA^{abc}[g^{\mu\nu}(k-p)^\mu + g^{\nu\rho}(p-q)^\mu + g^{\rho\mu}(q-k)^\nu]$$

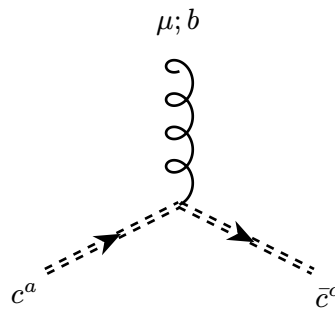
- Quads g:



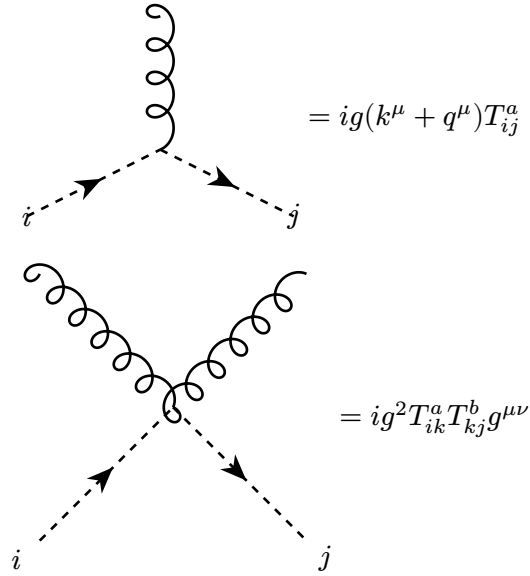
$$= -ig^2[f^{abe}f^{cde}(g^{\mu\rho}g^{\nu\sigma} - g^{\mu\sigma}g^{\nu\rho}) + f^{ace}f^{bde}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\sigma}g^{\nu\rho}) + f^{ade}f^{bce}(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma})]$$



$$= ig\gamma^\mu T_{ij}^a$$



$$= -gf^{abc}p^\mu$$



9.7 Helicity Spinors

Summary:

- Helicity spinors are a way to write polarizations ε_μ and momentum p_μ in terms of a more natural way (bispinors)
 - This is more convenient specifically for calculating gluon scattering
- They are written in the weyl representation where ψ_α and $\tilde{\chi}^{\dot{\alpha}}$ are left and right handed spinors
- Notice that spin 1 fields transform in the $(\frac{1}{2}, \frac{1}{2})$ representation of the Lorentz group
 - This entails that we can write them as bispinors instead of 4-vectors
- Remember the mathematical relations for spinors in the weyl representation:
 - $\varepsilon^{\alpha\beta}\psi_\alpha(x)\psi_\beta(x)$ and $\varepsilon_{\dot{\alpha}\dot{\beta}}\tilde{\psi}^{\dot{\alpha}}\tilde{\psi}^{\dot{\beta}}$ are L.I.
- Some useful relations with these (and the pauli matrices) are

$$g^{\mu\nu}\sigma_\mu^{\alpha\dot{\alpha}}\sigma_\nu^{\beta\dot{\beta}} = 2\varepsilon^{\alpha\beta}\varepsilon^{\dot{\alpha}\dot{\beta}}$$

$$\varepsilon_{\alpha\beta}\varepsilon_{\dot{\alpha}\dot{\beta}}\sigma^{\mu\dot{\beta}\beta} = \bar{\sigma}^\mu_{\dot{\alpha}\alpha}$$

- For the inner product, defined as $\psi\chi \equiv \varepsilon^{\alpha\beta}\psi_\alpha(x)\chi_\beta(x)$
 - $\psi\chi = \chi\psi$
- All of these spinors are fields. What happens when we use constant spinors?

Definition 9.7.1 (Helicity Spinor).

- $\mathbb{R}$ or $\mathbb{C}$ doublets that transform in the $(\frac{1}{2}, 0)$ or $(0, \frac{1}{2})$ representation of the Lorentz Group
- The inner products for these are
- $\langle \lambda \chi \rangle = \epsilon^{\alpha\beta} \lambda_\alpha \chi_\beta = \lambda_\alpha \chi^\alpha = -\lambda^\alpha \chi_\alpha$
- $[\lambda \chi] = \epsilon_{\alpha\beta} \tilde{\lambda}^\alpha \tilde{\chi}^\beta = \tilde{\lambda}^\alpha \tilde{\chi}_\alpha = -\tilde{\chi}_\alpha \tilde{\lambda}^\alpha$
- These products have some identities due to them commuting:

$$\begin{aligned}\langle \lambda \chi \rangle &= -\langle \chi \lambda \rangle, [\lambda \chi] = -[\chi \lambda] \\ [\lambda \lambda] &= \langle \lambda \lambda \rangle = 0\end{aligned}$$

9.7.1 Momentum

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- **Relations** with these spinors are:

$$\begin{aligned}p^{\alpha\dot{\alpha}} &\equiv \sigma_{\alpha\dot{\alpha}}^\mu p_\mu, \quad p_{\alpha\dot{\alpha}} = \bar{\sigma}_{\alpha\dot{\alpha}}^\mu p_\mu, \\ p^\mu &= \frac{1}{2} \bar{\sigma}^{\mu\alpha\dot{\alpha}} p_{\alpha\dot{\alpha}}, \quad p^\mu = \frac{1}{2} \bar{\sigma}_{\alpha\dot{\alpha}}^\mu p^{\alpha\dot{\alpha}}.\end{aligned}$$

- If $p^{\alpha\dot{\alpha}}$ is lightlike (that is, $\det(p^{\alpha\dot{\alpha}}) = 0$), we can express it in terms of an outer product

$$\begin{aligned}p^{\alpha\dot{\alpha}} &= \lambda^\alpha \tilde{\chi}^{\dot{\alpha}} \\ \lambda_\alpha &= \frac{z}{\sqrt{p^0 - p^3}} \begin{pmatrix} p^0 - p^3 \\ -p^1 - ip^2 \end{pmatrix}, \\ \tilde{\lambda}^{\dot{\alpha}} &= \frac{z^1}{\sqrt{p^0 - p^3}} (p^0 - p^3, -p^1 + ip^2)\end{aligned}$$

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- These momenta bispinors are invariant under $\lambda_\alpha \rightarrow z \lambda_\alpha$ and $\tilde{\lambda}^{\dot{\alpha}} \rightarrow \frac{1}{z} \tilde{\lambda}^{\dot{\alpha}}$
 - This is equivalent to a phase shift $z = e^{i\phi}$ if the momenta are real

9.7.1.2 Connecting Momentum and Helicity Spinors

- Notice that the dot product

$$\begin{aligned}
p \cdot q &= \frac{1}{4} g_{\mu\nu} \sigma_{\alpha\dot{\alpha}}^{\mu} \sigma_{\beta\dot{\beta}}^{\nu} \lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}} \chi^{\beta} \tilde{\chi}^{\dot{\beta}} \\
&= \frac{1}{2} \varepsilon_{\alpha\beta} \varepsilon_{\dot{\alpha}\dot{\beta}} \lambda^{\alpha} \tilde{\lambda}^{\dot{\alpha}} \chi^{\beta} \tilde{\chi}^{\dot{\beta}} = \frac{1}{2} \langle \chi \lambda \rangle [\chi \lambda]
\end{aligned}$$

- Therefore, we obtain that $\langle \chi \lambda \rangle$ and $[\chi \lambda]$ are the square roots of inner products up to a phase shift

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$$\begin{aligned}
\sum_i p_i &= 0 \\
\sum_j \lambda_{\alpha}^j \lambda_{\alpha}^j &= 0 \\
\sum_{j=1}^n j \rangle [j=1] [1+2] [2+\dots+n] [n=0]
\end{aligned}$$

- To help us with being terse, we write $i\rangle \equiv p_i\rangle$, etc.

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9.7.2 Polarizations

- All polarizations satisfy the relation that

$$\varepsilon_\mu^* \varepsilon^\mu = -1, \quad p_\mu \varepsilon^\mu = 0$$

- Notice that $\varepsilon_\mu \varepsilon^\mu = 0$

Example 9.7.2.1 (Helicity Polarizations).

- If the momentum is $p^\mu = (E, 0, 0, E)$ then we obtain polarizations

$$\epsilon_+^\mu = \frac{1}{\sqrt{2}}(0, 1, i, 0),$$

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- In order to write the polarizations in terms of the momentum, we need a reference momentum. In the case of gluon scattering which this is a very useful tool for, we take the reference momentum to be the momentum of the other gluons.
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$$[\varepsilon_p^+(v)]^{\alpha\dot{\alpha}} = \sqrt{2} \frac{r\rangle[p}{\langle rp\rangle}$$

$$\begin{aligned} \varepsilon_p^-(v) \cdot \varepsilon_p^+(v) &= \frac{1}{2} [\varepsilon_p^-(v)]_{\dot{\alpha}\alpha} [\varepsilon_p^+(v)]^{\alpha\dot{\alpha}} \\ &= \frac{1}{2} \frac{2}{[pr]\langle rp\rangle} \text{tr}\{r\rangle\langle pr\rangle[p] = -1 \end{aligned}$$

- since $\langle pp\rangle[pp] = 0$, we have that $\varepsilon^+ \varepsilon^+ = \varepsilon^- \cdot \varepsilon^- = 0$

10 Electroweak and SSB

Importance

- This is the 3rd and final force (outside of string theory) that is important in QFT.
- Important ideas from this section include *spontaneous symmetry breaking* and the higgs boson

10.1 Spontaneous Symmetry Breaking

10.1.1 Condmat/Discrete symmetries

- Spontaneous has some analogies to condensed matter physics which are worth noting
- For instance, suppose there was a sort of “critical temperature” in QFT, where after it, the mass² becomes negative.
- For instance, consider

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$$

- Notice that the minimum value of the potential is at $\phi = 0$ (important)
- We would like to define a ‘mass’ $\mu^2 \equiv -m^2$ for purposes of having the minimum of the potential being NOT $\phi = 0$

10.2 The Electroweak Force

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